Sharp ideal growth of metabelian restricted Lie algebras

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Abstract

We determine sharp ideal-growth terms and typical quotient structures for metabelian restricted Lie algebras over finite fields. For fixed generator rank $d \geq 3$, the logarithm to base q of the number of codimension-n restricted ideals is $n^2/4 + \gamma_d n^{2-1/d} + O(n^{2-2/d})$, with explicit γ_d. Rank two has the sharper formula $n^2/4 + n^{3/2}/2 - n \log_p n/2 + O(n)$, where p is the characteristic. These coarse formulas also hold for p-nilpotent quotients and for restricted isomorphism types. An exact bounded-rank compatibility mixture then yields finer laws under uniform ideal sampling. A typical quotient has a central invertible restricted-power part of dimension $n/2 + o(n)$; conditioned on whole-quotient p-nilpotence, its derived restricted part has one chain of length $n/2 + o(n)$, with classical Hall partition fluctuations. In rank two the proportion of whole-p-nilpotent quotients is $q^{-n/2+O(\sqrt{n})}$. The proof combines prescribed-head enumeration, module boundary estimates, refined-flag counting, and an exact central torus convolution. For homogeneous metabelian sources, Krull dimension and Hilbert multiplicity determine the first growth correction; graph commutation relations encode vertex connectivity and weighted minimum cuts.

Keywords. Restricted Lie algebra; ideal growth; metabelian Lie algebra; finite field; asymptotic enumeration; typical quotient; nilpotence probability; graph commutation.

Contents

1 Introduction **3**
1.1 Module dimension and graph cuts . . . 3
1.2 Typical structures and a rare-event exponent . . . 4
1.3 Proof mechanism and classical inputs . . . 4

2 Restricted hulls and prescribed heads **5**
2.1 The hull decomposition . . . 5
2.2 Exact head enumeration . . . 6
2.3 A lower construction at every large dimension . . . 6

3 Boundary rank and relation-module counts **7**
3.1 Relation-module entropy . . . 10
3.2 Mixed Hom estimates . . . 10
3.3 Fixed-rank modules . . . 11
3.4 Counting points of positive strata . . . 11

4 The free growth formula **12**

5 The rank-two logarithmic term **13**

6 Homogeneous quotients and graph applications **14**
6.1 Classical cut data and the growth correction . . . 14
6.2 The path on four vertices . . . 15
6.3 Scope of the comparison . . . 15

7 Typical layers and prescribed-head laws **16**
7.1 Notation and measures . . . 16
7.2 Concentration of the quotient layers . . . 16
7.3 Exact nilpotent heads and the Fitting law . . . 18

8 Outer compatibility and global partition laws **20**
8.1 The exact conditional outer count . . . 21
8.2 Uniform prescribed-head inputs . . . 22
8.3 Global laws under uniform ideal counting . . . 23
8.4 The global nilpotent ensemble has one giant chain . . . 24

9 Central invertible factors and adjacent growth **26**
9.1 Two different forms of nilpotence . . . 26
9.2 An exact marked convolution, and a type identity . . . 27
9.3 Adjacent-dimension smoothness . . . 28
9.4 The outer central torus has an exponentially small cost . . . 28
9.5 Factoring the whole-quotient probability . . . 29

10 The rank-two whole-quotient nilpotence rate **29**
10.1 Statement and notation . . . 29
10.2 Near-extremal planar relation ideals have sublinear entropy . . . 30
10.3 The effective outer penalty . . . 31
10.4 A primary comparison model with the same h, g . . . 32
10.5 Proof of the locality cost and the full rate . . . 32

11 Further questions and reproducibility **33**
11.1 Beyond rank two . . . 33
11.2 Exact controls . . . 33

1 Introduction

The number of finite-dimensional quotients measures the freedom left by the defining identities of a Lie algebra. For finitely generated metabelian restricted Lie algebras, this freedom is already quadratic in the logarithm of the index. The main issue is to identify its leading constant and then the finer scales that distinguish different source algebras.

Fix a finite field $\mathbb{F}_q$ of characteristic p. A restricted ideal is a Lie ideal closed under the restricted p-map. Write F_d for the free metabelian restricted Lie algebra on d generators and $c_n(F_d)$ for its number of restricted ideals of $\mathbb{F}_q$-codimension n. Both d and q remain fixed as $n \to \infty$. Petrogradsky and Subbotin established quadratic logarithmic growth and proposed the leading asymptotic $\log_q c_n(F_d) \sim n^2/4$ [1]. Their construction separates the ordinary commutator module from its central positive restricted powers. We refine the count at precisely this interface: the image of an ideal in the first restricted layer must be held fixed during the remaining enumeration.

There are four counting sequences in the statements below: c_n, the number of restricted ideals; c_n^{nil}, the number whose quotient is p-nilpotent; and t_n, t_n^{nil}, the respective numbers of restricted $\mathbb{F}_q$-Lie isomorphism types of quotients. Here p-nilpotent means that every element is killed by an iterated restricted p-power. For F_d, the type counts include exactly the algebras generated by at most d elements in this restricted variety. No bound on restricted height is imposed.

Theorem 1.1 (Fixed-rank growth)**.** *For fixed $d \geq 3$ and q, each of the four sequences A_n associated with F_d satisfies*

$$\log_q A_n = \frac{n^2}{4} + \gamma_d n^{2-1/d} + O_{d,q}(n^{2-2/d}), \qquad \gamma_d = \frac{d}{2^{2-1/d}} \left(\frac{d-1}{d!} \right)^{1/d}.$$

The case of two generators has an additional visible scale.

Theorem 1.2 (Rank-two three-term growth)**.** *For each of the four sequences associated with F_2,*

$$\log_q A_n = \frac{n^2}{4} + \frac{1}{2} n^{3/2} - \frac{1}{2} n \log_p n + O_q(n).$$

In particular, the leading constant is $1/4$ in every fixed rank $d \geq 2$. The base p of the logarithm in the third term is essential. It reflects a restricted-PBW constraint on the dimension of the abelian outer quotient; the positive term instead measures the boundary of the ordinary relation module.

1.1 Module dimension and graph cuts

The counting mechanism extends to a fixed homogeneous quotient G of an ordinary free metabelian Lie algebra, with degree-one generators $x_1, \ldots, x_d$ and no degree-one relations. Put $M = G'$ and let $R = \mathbb{F}_q[X_1, \ldots, X_d]$ act by the commuting generator adjoints. Let L_G be the universal restricted hull of G; its construction inside $U(G)$ is recalled in Section 2.1. If M has Krull dimension δ and standard Hilbert multiplicity e, our normalization is

$$\dim_{\mathbb{F}_q} M/\mathfrak{m}^s M = \frac{e}{\delta!} s^\delta + O_G(s^{\delta-1}).$$

Theorem 1.3 (Dimension and multiplicity)**.** *If $\delta \geq 2$, the four quotient-count sequences of L_G satisfy*

$$\log_q A_n = \frac{n^2}{4} + \frac{\delta (e/\delta!)^{1/\delta}}{2^{2-1/\delta}} n^{2-1/\delta} + O_{G,q}(n^{2-2/\delta} + n \log n).$$

For type counts, a type is admitted when it has an epimorphism from this fixed L_G.

For graph commutation relations, the module invariants are combinatorial. Let Γ be a noncomplete graph on d vertices and set

$$\delta_\Gamma = \max\{|W| : c(\Gamma_W) \geq 2\}, \qquad e_\Gamma = \sum_{|W|=\delta_\Gamma} (c(\Gamma_W) - 1),$$

where Γ_W is the induced subgraph and c counts its components.

Corollary 1.4 (Graph quotient growth)**.** *For the restricted metabelian algebra defined by $[x_i, x_j] = 0$ on the edges of Γ, the formula of Theorem 1.3 holds with $(\delta, e) = (\delta_\Gamma, e_\Gamma)$. If Γ is connected, then $\delta_\Gamma = d - \kappa(\Gamma)$, where κ is vertex connectivity.*

The correction exponent therefore recovers vertex connectivity when the vertex number is known. Its coefficient recovers the weighted number of minimum vertex cuts. The cut polynomial and its relation to the metabelian Hilbert series are due to Papadima–Suciu [7]; the application here is to finite-codimension restricted ideals.

1.2 Typical structures and a rare-event exponent

The two counting sequences have the same first growth scales but support different typical restricted-power structures. Write $Q = L/H$ and $C = \langle Q' \rangle_p$. Under uniform ideal sampling, the invertible Fitting part of C is a central restricted ideal of dimension $n/2 + o_{\mathrm{Pr}}(n)$. Under sampling conditioned on Q being p-nilpotent, C instead has one restricted-power chain of length $n/2 + o_{\mathrm{Pr}}(n)$. The remaining finite partitions follow classical Hall/Cohen–Lenstra laws. The exact compatibility mixture in Theorem 8.1 is what permits these laws to pass from prescribed heads to the actual ideals.

For the two-generator free source we determine the finer ratio

$$\frac{c_n^{\mathrm{nil}}(F_2)}{c_n(F_2)} = q^{-n/2 + O_q(\sqrt{n})}.$$

The leading and logarithmic terms of the two sequences agree, yet p-nilpotence costs a precise linear exponent. Theorem 10.1 proves this by combining a finite-characteristic refined-flag bound with a comparison that preserves the relation ideal's colength and generator number. The probability measure here is uniform on ideals.

In characteristic two, height-two restricted models also produce the quadratic orbit equation $f \circ f = f + \mathrm{id}$, studied in the companion work [12]. The present sources allow unrestricted power height. Their prescribed-head and central-extension interfaces connect the two studies, while the fixed-rank, growing-dimension enumeration here has its own limiting regime.

1.3 Proof mechanism and classical inputs

The first step counts submodules of a free abelian restricted algebra with a prescribed image modulo its positive powers. Subspace-lattice Möbius inversion gives an exact generating series and a bound uniform in the number of generators. For a relation module of colength h, the relevant number of free restricted generators is $h + g$, where g is its boundary generator number. Its contribution is approximately $q^{(h+g)(b-h)}$ in inner quotient dimension b. The quadratic maximum occurs near $h = b/2$; the sharp boundary coefficient then gives the first correction.

Berman's generator extremum [2] supplies the rank-one boundary constant. We pass to the actual rank of a fixed module and retain the length allocation to its lower-dimensional quotient. Briançon–Iarrobino [3] already proved the classical $h^{2-2/d}$ tangent-space order in every characteristic. We use mixed Hom spaces to control fixed-rank modules, and the positive

grading of Gröbner strata [6] to obtain genuine finite-field point counts. The two-dimensional Hom calculation is also recorded in [5, Proposition 4.1]. Segal's planar ideal-growth theorem [4] gives the stronger linear-exponential relation count needed for the $O(n)$ error in rank two.

Thus the proof uses established restricted-hull, generator-boundary and Hilbert-scheme tools. The precise conclusions concern their combination in restricted ideal enumeration: the coefficient γ_d, the rank-two logarithmic loss, and the transfer to module dimension and multiplicity. Section 2 gives the exact count, Section 3 the module estimates, Sections 4–5 prove the free formulas, and Section 6 proves the transfer and graph application.

Sections 7–8 pass from the exact counts to typical layers and global partition laws. Section 9 separates ordinary nilpotence from restricted-power nilpotence through a canonical splitting and adjacent-growth estimates. Section 10 completes the whole-quotient rarity exponent in rank two.

2 Restricted hulls and prescribed heads

2.1 The hull decomposition

Let G be as in Theorem 1.3. Embed G in $U(G)$ by PBW and take its restricted closure, with $x^{[p]} = x^p$. If a basis of G consists of the head vectors x_i and a basis m_λ of $M = G'$, PBW makes their individual p^j-powers linearly independent. Their span is closed under brackets and the restricted power: restricted adjoints reduce brackets to ordinary iterated brackets, and Jacobson's sum formula adds Lie words. Consequently these powers form a basis of L_G. The universal property follows by extending a Lie map $G \to B$ to $U(G) \to u(B)$ and restricting to this closure.

As M is abelian, $(\operatorname{ad} m)^2 = 0$ on G for $m \in M$. Hence its positive restricted powers are central. In particular

$$L_G = T \oplus Z, \quad T = \operatorname{span}_{\mathbb{F}_q}\{x_i^{[p^j]} : i \le d, j \ge 0\}, \quad Z = \langle M \rangle_p = M \oplus Z^{[p]}, \quad Z^{[p]} \subseteq Z(L_G). \tag{1}$$

Here T is a vector-space complement and Z a restricted ideal; Z is the free abelian restricted hull of the vector space M. The quotient L_G/Z is free abelian restricted of rank d. These are the classical ingredients of [1, §4].

Let H be a restricted ideal of codimension n. Define

$$B = H \cap Z, \quad b = \dim Z/B, \quad a = n - b, \quad N = (B + Z^{[p]})/Z^{[p]} \subset M, \quad h = \dim M/N. \tag{2}$$

Then $0 \le h \le b$ and N is an R-submodule. Since brackets ignore the $Z^{[p]}$ part,

$$[L_G, B] = \mathfrak{m}N \subset B, \qquad C_N = \langle \mathfrak{m}N \rangle_p \subset B.$$

The free abelian restricted algebra Z/C_N has rank

$$S = \dim M/\mathfrak{m}N = h + g(N), \qquad g(N) = \dim N/\mathfrak{m}N. \tag{3}$$

The head image of B/C_N is the fixed space $N/\mathfrak{m}N$ of codimension h in $M/\mathfrak{m}N$.

For later use we recall the remaining counting bound. There are $O_{d,q}(q^{da})$ possibilities for the outer ideal $(H + Z)/Z$. For fixed outer ideal and B, there are at most q^{db} lifts: an outer ideal is free of rank d as a module over the skew polynomial ring, and a lift is determined by the images of these d generators in Z/B. Thus the outer choices and lifts together cost at most $O_{d,q}(q^{dn})$, as in [1, (13)–(14)]. This argument uses only (1), so it also applies to L_G.

2.2 Exact head enumeration

Put $\Omega = \mathbb{F}_q[t;\sigma]$, $\sigma(c) = c^p$, with t acting as the restricted power. All modules here are left modules. Let $u_S(b,h)$ count colength-b submodules of Ω^S whose head image in $\Omega^S/t\Omega^S$ is one fixed codimension-h space. Write $\Theta_q = \prod_{j\geq 1}(1-q^{-j})^{-1}$.

Proposition 2.1 (Prescribed-head series)**.** *For $0 \leq h \leq S$,*

$$\sum_{b\geq 0} u_S(b,h) z^b = z^h \frac{\prod_{i=0}^{S-h-1}(1-q^i z)}{\prod_{i=1}^{S}(1-q^i z)}, \qquad u_S(b,h) \leq \Theta_q q^{S(b-h)} \quad (b \geq h).$$

For $S \geq 1$, one also has $u_S(b,h) \geq (1-q^{-S})q^{S(b-h)}$.

Proof. The total colength series of a free rank-S module is $A_S(z) = \prod_{i=1}^{S}(1-q^i z)^{-1}$ [1, Theorem 6]. One can also derive it by triangular generators: projection to one free coordinate gives a monic polynomial of degree v, with q^v choices, and lifts modulo a colength-u submodule of the remaining coordinates have q^u choices. Hence $A_S(z) = (1-qz)^{-1}A_{S-1}(qz)$. For a head space of codimension j, its inverse image is free of rank S and colength j, using adapted generators e_i, te_j. Möbius inversion over subspaces of the prescribed head gives

$$u_S(b,h) = \sum_{j=0}^{S-h} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} S-h \\ j \end{bmatrix}_q [z^{b-h-j}] A_S(z).$$

The finite q-binomial theorem gives the series. If $h < S$, cancellation reduces it to $z^h(1-z)\prod_{i=S-h}^{S}(1-q^i z)^{-1}$. The degree-$m$ denominator coefficient D_m, $m = b-h$, satisfies $D_m \geq q^S D_{m-1}$ and $D_m \geq q^{Sm}$, where $D_{-1} = 0$. Thus its consecutive difference has the asserted lower bound. For the upper bound, normalize the largest geometric factor by q^{Sm} and sum the smaller ones; their product is at most $\prod_{j=1}^{h}(1-q^{-j})^{-1} \leq \Theta_q$. When $h = S$, use the total series shifted by S. For $S = 0$ only $b = h = 0$ occurs. □

Lemma 2.2 (A nilpotent family with the same head)**.** *If $b > h \geq 1$, the kernels having quotient block partition $(b-h+1, 1^{h-1})$ and the prescribed head number exactly*

$$q^{(S-1)(b-h)} \frac{1-q^{-h}}{1-q^{-1}}.$$

Proof. Put $m = b-h$ and call the quotient Q. A head-isomorphism $\mathbb{F}_q^S/V \to Q/tQ$ has $|\operatorname{GL}_h(q)|$ choices. Lifting all generator images through tQ gives q^{Sm} choices. They are surjective, since a nilpotent cokernel with zero head is zero. An endomorphism of Q is determined by its long-chain top (b parameters) and its $h-1$ short-chain tops in $\ker t$ ($h(h-1)$ parameters). Thus the endomorphism space has size q^{h^2+m}. On the head, the long scalar must be nonzero and the short block invertible; the automorphism count is $q^{h^2+m}\phi_{h-1}\phi_1$, where $\phi_j = \prod_{i=1}^{j}(1-q^{-i})$. Division by this free target-automorphism action gives the formula. Images down a chain are successive semilinear powers, so the argument applies to the genuine Frobenius twist as well. □

2.3 A lower construction at every large dimension

Suppose M is generated in ordinary degree two. Set

$$N_s = \mathfrak{m}^s M,\ C_s = \langle \mathfrak{m}^{s+1} M \rangle_p, \quad h_s = \dim M/N_s,\ g_s = \dim N_s/\mathfrak{m}N_s,\ S_s = h_s + g_s.$$

For $r_s = \lceil \log_p(s+2) \rceil$ and $b = n - dr_s > h_s \geq 1$, choose a submodule B/C_s with prescribed head $N_s/\mathfrak{m}N_s$ using Lemma 2.2. Define

$$H_B = B \oplus \operatorname{span}_{\mathbb{F}_q}\{x_i^{[p^j]} : i \leq d,\ j \geq r_s\}. \tag{4}$$

This is a restricted ideal of codimension n. Indeed, $[L_G, B] \subseteq \mathfrak{m}^{s+1}M$ and

$$[x_i^{[p^u]}, x_j^{[p^v]}] = X_i^{p^u-1} X_j^{p^v-1}[x_i, x_j]$$

up to the harmless bracket-orientation sign. All brackets involving a discarded tower lie in $\mathfrak{m}^{p^{r_s}-1}M \subseteq C_s$. Modulo C_s the sum in (4) is central and closed under restricted powers. The vector decomposition proves $H_B \cap Z = B$ and exact codimension $dr_s + b$. The inner quotient is nilpotent and the outer towers have height at most r_s, so the whole quotient is p-nilpotent. Distinct B give distinct ideals. Consequently

$$\log_q c_n^{\mathrm{nil}}(L_G) \geq (S_s - 1)(n - dr_s - h_s) + O_q(1). \tag{5}$$

Finally a fixed quotient type has at most q^{dn} marked epimorphisms from L_G. Hence in either quotient class

$$q^{-dn} c_n \leq t_n \leq c_n. \tag{6}$$

This transfers all terms larger than linear from ideals to types.

3 Boundary rank and relation-module counts

Let $R = \mathbb{F}_q[X_1, \ldots, X_D]$, $D \geq 2$, and $\mathfrak{m} = (X_1, \ldots, X_D)$. All modules in this section are finitely generated. Put $\alpha = 1 - 1/D$, $\eta = 1 - 2/D$ and $C_D = D/(D!)^{1/D}$.

Proposition 3.1 (Sharp fixed-module boundary)**.** *If a fixed R-module M has positive rank ρ, every colength-h submodule satisfies*

$$\dim N/\mathfrak{m}N \leq C_D \rho^{1/D} h^{1-1/D} + O_M((h+1)^{1-2/D}).$$

We give the proof with the polynomial-ring dimension denoted by d in the calculations below; thus $d = D$, $\eta = 1-2/d$ and $\alpha = 1-1/d$. The rank-one input is Berman's extremum [2]; Segal's local generator estimate [4, Proposition 3.3] controls the fixed torsion quotient. In this proof k denotes the global colength, h its length after localizing at $\mathfrak{m}$, and $\mu_{\mathfrak{m}}(U) = \dim U/\mathfrak{m}U$.

Proof of Proposition 3.1.

Step 1: a quantitative rank-one consequence of Berman. Let $A = \mathbb{F}_q[[X_1, \ldots, X_d]]$, and let $I \subset A$ have colength h. Berman's result and its associated-graded reduction give

$$\mu_A(I) \leq C_d h^{\alpha} + O_d((h+1)^{\eta}).$$

The error term follows from the comparison of adjacent powers in the extremal family. The colength and generator count of $\mathfrak{m}^j$ are

$$L_j = \binom{j+d-1}{d}, \qquad G_j = \binom{j+d-1}{d-1}.$$

If $L_j \leq h < L_{j+1}$, Berman's extremal ideal has at most G_{j+1} generators. Now

$$L_j = \frac{j^d}{d!} + O_d(j^{d-1}), \qquad G_{j+1} = \frac{j^{d-1}}{(d-1)!} + O_d(j^{d-2}).$$

Since $h \geq L_j$ and $C_d/(d!)^{\alpha} = 1/(d-1)!$, this gives the rank-one bound. The finitely many small h, including $h = 0$ where $I = A$, are absorbed by the $(h+1)^{\eta}$ term. Completion preserves length and generator number, so the rank-one bound also applies in the regular local ring $R_{\mathfrak{m}}$.

Step 2: free rank ρ contributes $\rho^{1/d}$. Let $K \subset A^\rho$ have colength h. Project to one coordinate, and denote the image ideal by I and the kernel submodule in $A^{\rho-1}$ by K'. The exact sequences

$$0 \to K' \to K \to I \to 0, \qquad 0 \to A^{\rho-1}/K' \to A^\rho/K \to A/I \to 0$$

give both the generator subadditivity and additive colengths. Iterating yields colengths $h_1, \dots, h_\rho \geq 0$ with $\sum h_i = h$ and

$$\mu_A(K) \leq \sum_{i=1}^{\rho} \left(C_d h_i^\alpha + O_d((h_i+1)^\eta)\right).$$

Concavity gives $\sum h_i^\alpha \leq \rho^{1/d} h^\alpha$. As ρ is fixed, the error sum is $O_{d,\rho}((h+1)^\eta)$. Thus

$$\mu_A(K) \leq C_d \rho^{1/d} h^\alpha + O_{d,\rho}((h+1)^\eta).$$

This argument uses the actual rank ρ. Pulling back under an arbitrary presentation $R^r \twoheadrightarrow M$ and applying the free-rank bound with that r would give only the weaker coefficient $C_d r^{1/d}$.

Step 3: isolate the lower-dimensional quotient. Choose ρ elements of M independent over the fraction field. They span a fixed free submodule $F \simeq R^\rho$. The quotient $T = M/F$ is a fixed torsion module, hence has Krull dimension at most $d-1$. For a submodule N set

$$K = N \cap F, \qquad U = (N+F)/F \subset T.$$

There are exact sequences

$$0 \to K \to N \to U \to 0, \qquad 0 \to F/K \to M/N \to T/U \to 0.$$

Localize at $\mathfrak{m}$, and let h_F, h_T be the two finite lengths on the right of the two exact sequences. They satisfy

$$h_F + h_T = h \leq k.$$

Indeed localization selects the $\mathfrak{m}$-primary part of the finite global quotient, and its residue field is $\mathbb{F}_q$. Thus this step does not require the original quotient to be $\mathfrak{m}$-primary or homogeneous.

Put $\beta = (d-2)/(d-1)$. Segal's Proposition 3.3 applied to the fixed local module $T_\mathfrak{m}$ gives

$$\mu_\mathfrak{m}(U) \leq D_M (h_T+1)^\beta.$$

If its dimension is smaller than $d-1$, its exponent is no larger; if its dimension is zero, its submodules have bounded generator number, so the torsion estimate still holds. Right exactness after tensoring the first sequence in the two exact sequences with the residue field, followed by the free-rank bound, gives

$$\mu_\mathfrak{m}(N) \leq C_d \rho^{1/d} h_F^\alpha + D_M (h_T+1)^\beta + O_M((h_F+1)^\eta).$$

Step 4: keep the length allocation to sharpen the error. Replacing both h_F, h_T by k too early would only give an error $O(k^\beta)$. Preserve $h_F + h_T = h$ instead. For $h \geq 1$, concavity at h gives, with $C = C_d \rho^{1/d}$ and $t = h_T$,

$$C(h-t)^\alpha \leq C h^\alpha - C\alpha h^{-1/d} t.$$

For $d > 2$, the elementary maximum

$$\sup_{t \geq 0} \{D(t+1)^\beta - C\alpha h^{-1/d} t\} = O_{C,D,d}(h^{\beta/[d(1-\beta)]} + 1) = O_{C,D,d}(h^{1-2/d} + 1)$$

follows by maximizing $Dt^\beta - ch^{-1/d}t$ and using $(t+1)^\beta \leq t^\beta + 1$. For $d = 2$, $\beta = \eta = 0$ and the bound is simply $O(1)$. Insert the allocation estimate into the combined estimate and use $h \leq k$ to obtain Proposition 3.1. The case $h = 0$ has bounded generator number and is absorbed by the same error. This completes the general fixed-module proof.

An explicit rank-$d-1$ free part in the commutator module. Let now $M = L_0'$ for the ordinary free metabelian Lie algebra on $x_1, \dots, x_d$. It is generated over R by $[x_i, x_j]$. Take

$$F = \bigoplus_{j=2}^{d} R[x_1, x_j].$$

The displayed generators are independent. One direct witness is the metabelian semidirect product of the abelian space with basis $v_1, \dots, v_d$ and the abelian R-module R^d, where v_i acts as multiplication by X_i. Map x_i to (v_i, e_i). The image of $[x_1, x_j]$ is $X_1 e_j - X_j e_1$. Its jth coordinate shows that a relation $\sum_{j\geq 2} a_j[x_1, x_j] = 0$ forces $X_1 a_j = 0$, hence $a_j = 0$.

Jacobi gives

$$X_1[x_i, x_j] = X_i[x_1, x_j] - X_j[x_1, x_i] \in F.$$

Therefore $X_1 M \subset F$, and

$$T = M/F \quad \text{is a fixed finite module over} \quad R' = \mathbb{F}_q[X_2, \dots, X_d].$$

In particular $\operatorname{rank}_R M = d - 1$. No coefficient $\binom{d}{2}$ from the usual commutator presentation survives in the leading term.

For this particular T, there is a short elementary alternative to the local torsion step. It uses *global* colengths $k_F = \dim F/K$ and $k_T = \dim T/U$, with $k_F + k_T = k$. Under a fixed presentation $(R')^r \twoheadrightarrow T$, pull back U. Its colength is exactly k_T. For a reduced module Groebner basis, each component's standard monomials are a finite downset. In $e = d - 1$ variables, the minimal excluded monomials of a downset of size j form an antichain of size $O_e((j+1)^{1-1/e})$: those in a box of side a number at most a^{e-1}, while those with one coordinate at least a use disjoint predecessor chains of length a, giving at most ej/a more. Choose $a \asymp (j+1)^{1/e}$. For $e = 1$ there is at most one minimal excluded monomial in each component. The resulting Groebner generators map to global generators of U. Thus they give $\mu_{\mathfrak{m}}(U) \leq D(k_T + 1)^{\beta}$; this is a global-$k_T$ bound and does not by itself replace k_T by the local length h_T in the torsion estimate. For K, localization has colength at most k_F, so the free-rank bound gives $\mu_{\mathfrak{m}}(K) \leq C k_F^{\alpha} + O((k_F + 1)^{\eta})$. Consequently

$$\mu_{\mathfrak{m}}(N) \leq C(k - k_T)^{\alpha} + D(k_T + 1)^{\beta} + O((k+1)^{\eta}).$$

Apply exactly the optimization the allocation estimate, now with total length k and allocation $t = k_T$. This proves the commutator-module version of Proposition 3.1 using only Berman's sharp rank-one theorem plus the elementary free-module, Jacobi and downset arguments above.

Sharpness from the exact commutator Hilbert function. For $N_s = \mathfrak{m}^s M$, the standard metabelian basis gives

$$\dim M_\ell = (\ell - 1)\binom{d + \ell - 2}{d - 2}.$$

Writing $A_d = (d-1)/d!$, one consequently has

$$k_s = \dim M/N_s = A_d s^d + O_d(s^{d-1}), \qquad g_s = \mu_{\mathfrak{m}}(N_s) = d A_d s^{d-1} + O_d(s^{d-2}).$$

It follows that

$$\frac{g_s}{k_s^{1-1/d}} \longrightarrow d A_d^{1/d}.$$

This proves sharpness of the commutator boundary coefficient along a sequence of actual submodules of the fixed module M.

3.1 Relation-module entropy

Proposition 3.2 (Fixed-rank submodule count)**.** *For a fixed finitely generated module over $\mathbb{F}_q[X_1, \dots, X_D]$, the number a_h of colength-h submodules satisfies*

$$\log_q a_h = O_{M,q}((h+1)^{2-2/D} + h \log(h+1)).$$

For $D \geq 3$ the second summand is absorbed by the first.

The diagonal tangent-space order is classical [3]. We include the mixed estimate and the finite-field point argument needed for fixed-rank modules. In the following proof d denotes D.

3.2 Mixed Hom estimates

For monomial finite-colength ideals $I, J \subset R_d$ write $a = \dim R_d/I$, $b = \dim R_d/J$. We prove the bound

$$\dim_K \operatorname{Hom}_{R_d}(I, R_d/J) \leq C_d\big((a+1)(b+1)^{1-2/d} + (b+1)(a+1)^{1-2/d}\big).$$

The harmless $+1$ handles $I = R_d$ and $J = R_d$.

Characteristic independence and simultaneous degeneration. Use monomial generators of I and the standard monomials of R_d/J. The pairwise Taylor syzygies impose only coefficient equations of the form $u = v$ or $u = 0$: multiplication by a monomial sends distinct standard monomials to distinct monomials until they vanish in J. Thus the dimension of this Hom space is independent of K. It is enough to bound it over $\mathbb{Q}$.

Over an infinite characteristic-zero field, choose the same generic linear change of variables for I and J and degenerate both with the same term order. Their two generic initial ideals are strongly stable. The Hom dimension is upper semicontinuous in this simultaneous flat degeneration. One direct justification is to take finite presentations of the source ideal over the one-parameter base. The source ideal is flat because it is the kernel of a flat family of finite quotients of the polynomial ring. The presentation remains a presentation on every fiber. Applying Hom into the flat, finite-rank target gives the kernel of a matrix between finite free modules over the parameter ring. Kernel dimension can only increase at specialization. Consequently it suffices to prove the mixed estimate for strongly stable I, J. Only homogeneous monomial ideals are being moved in this step, so no inhomogeneous version of generic initial ideals is required.

Base case $d = 2$. The classical two-dimensional regular-local formula gives

$$\dim \operatorname{Hom}(I, R_2/J) = b + \dim \operatorname{Ext}^2(R_2/I, R_2/J) \leq a + b.$$

See also [5, Proposition 4.1]. Their proof uses the Euler characteristic and the right-exact equality $\operatorname{Ext}^2(R_2/I, R_2/J) = \operatorname{Ext}^2(R_2/I, R_2) \otimes R_2/J$; the canonical module $\operatorname{Ext}^2(R_2/I, R_2)$ has length a. For monomial ideals all finite support is at the origin, so localization does not change any dimension. The case $I = R_2$ is immediate.

Induction: retain the mixed powers. Let $d \geq 3$, let $A = K[x_2, \dots, x_d]$, and write

$$I = \bigoplus_{i \geq 0} x_1^i I_i, \qquad J = \bigoplus_{j \geq 0} x_1^j J_j.$$

Set $r_0 = \min\{i : I_i = A\}$ and $s_0 = \min\{j : J_j = A\}$. These include $r_0 = 0$ or $s_0 = 0$ for the unit ideal. Strong stability and $x_1^{r_0-1} \notin I$ imply that every monomial of degree at most $r_0 - 1$ is standard. Thus

$$a \geq \binom{r_0 + d - 1}{d} \geq r_0^d/d!, \qquad r_0 + 1 \leq C_d(a+1)^{1/d},$$

and similarly $s_0 \leq C_d(b+1)^{1/d}$. Put $a_i = \dim A/I_i$, $b_j = \dim A/J_j$. Then $\sum_{i=0}^{r_0} a_i = a$, $a_{r_0} = 0$, and $\sum_{j=0}^{s_0-1} b_j = b$.

An R_d-linear map is determined by its restrictions to $x_1^i I_i$ for $0 \leq i \leq r_0$, since every later slice is generated from $x_1^p A$ by multiplication by x_1. Projecting those restrictions onto the s_0 target slices gives an injective linear map

$$\operatorname{Hom}_{R_d}(I, R_d/J) \hookrightarrow \bigoplus_{i=0}^{r_0} \bigoplus_{j=0}^{s_0-1} \operatorname{Hom}_A(I_i, A/J_j).$$

The omitted compatibility equations only decrease its dimension. No equality of source and target ideals is assumed.

Use the inductive mixed estimate and put $\theta = 1 - 2/(d-1) \in [0, 1)$. The double sum is bounded by a constant times

$$\left(\sum_i (a_i+1)\right)\left(\sum_j (b_j+1)^\theta\right) + \left(\sum_j (b_j+1)\right)\left(\sum_i (a_i+1)^\theta\right).$$

For $s_0 > 0$, Jensen's inequality and the small x_1-power bound give

$$\sum_j (b_j+1)^\theta \leq s_0^{1-\theta}(b+s_0)^\theta \leq C_d(b+1)^{\theta+(1-\theta)/d} = C_d(b+1)^{1-2/d}.$$

Also $\sum_j (b_j+1) = b + s_0 \leq C_d(b+1)$. Identical estimates, with r_0+1 slices, hold for a_i. When $s_0 = 0$ the Hom space is zero. This proves the mixed estimate for every d. Replacing the mixed estimate by a single convex power before summing would give a weaker harmonic-exponent recurrence; the concavity of the separate power is what avoids that loss.

3.3 Fixed-rank modules

A monomial finite-colength submodule of R_d^r has the form $J = \bigoplus_{i=1}^r I_i e_i$. Put $h_i = \dim R_d/I_i$ and $\sum h_i = h$. Then

$$\operatorname{Hom}_{R_d}(J, R_d^r/J) = \bigoplus_{i,j} \operatorname{Hom}_{R_d}(I_i, R_d/I_j).$$

Since r is fixed, the mixed estimate gives

$$\dim \operatorname{Hom}_{R_d}(J, R_d^r/J) = O_{d,r}((h+1)^{2-2/d}).$$

3.4 Counting points of positive strata

Fix a degree-compatible monomial module order. For every monomial initial submodule J, the monic reduced module-Groebner basis has the form

$$x^\alpha e_i - \sum_{x^\beta e_j \text{ standard and smaller}} c_{\alpha i,\beta j} x^\beta e_j.$$

Buchberger remainders give a finite coefficient ring defining the stratum with initial module J. Its $\mathbb{F}_q$-points correspond bijectively to the submodules being counted. Assign positive integer weights to the finitely many coefficient variables by representing the finite leading-versus-tail comparisons with a sufficiently separated monomial weight vector and component weights. The Buchberger equations are homogeneous for these weights. Hence the coefficient ring is a connected positively graded algebra.

For such an algebra, homogeneous lifts of a basis of the cotangent space at the origin generate the entire algebra, by induction on the positive weight. It therefore admits a closed immersion into affine space whose dimension is that cotangent dimension. This is the classical elimination property for positively graded Groebner strata; see [6, Lemma 3.2 and Proposition 3.4]. The module version follows by the same monic reduction and positive-weight argument.

The tangent space of the stratum at J injects into $\mathrm{Hom}_{R_d}(J, R_d^r/J)$: a first-order change sends each monic generator to its first-order standard tail, and the syzygies are exactly the condition that this defines a homomorphism. Thus the tangent estimate bounds the cotangent dimension, and the stratum has at most $q^{C_{d,r}(h+1)^{2-2/d}}$ rational points. This controls rational points directly; no bound on component degrees or number of components is being suppressed in a geometric-dimension estimate.

Every standard monomial has each coordinate exponent less than h, since its divisor chain in that coordinate is standard. Consequently there are at most $(r(h+1)^d)^h$ possible sets of h standard module monomials, even without imposing downset closure. Multiplying this bound by the bound for each stratum proves the submodule-count bound.

Finally, for any fixed finitely generated R_d-module M, pullback under a fixed presentation $R_d^r \twoheadrightarrow M$ injects its colength-h submodules into those of R_d^r with the same colength. Therefore the submodule-count bound applies to the relation module used in the restricted-Lie growth problem, independently of its presentation rank.
This completes the proof of Proposition 3.2.

4 The free growth formula

For the ordinary free metabelian commutator module on d points, $\mathrm{rank}_R M = d-1$. An explicit free submodule is $\bigoplus_{j=2}^{d} R[x_1, x_j]$; Jacobi makes its quotient annihilated by X_1. Proposition 3.1 gives

$$g(N) \le Ch^\alpha + O_d((h+1)^{1-2/d}), \quad C = d((d-1)/d!)^{1/d}, \quad \alpha = 1 - 1/d. \tag{7}$$

Proof of Theorem 1.1. At fixed (b,h), Proposition 2.1 and (3) bound the inner choices by $\Theta_q q^{(h+g(N))(b-h)}$. The relation-module count in Proposition 3.2, the q^{dn} outer and lifting factor, and the sum over at most $(n+1)^2$ parameter pairs all fit in the error $O_{d,q}(n^{2-2/d})$. Thus

$$\log_q c_n(F_d) \le \max_{0\le h\le b\le n} \{h(b-h) + Ch^\alpha(b-h)\} + O_{d,q}(n^{2-2/d}).$$

Increase b to n. Outside $[n/4, 3n/4]$, the quadratic loses at least $n^2/16$, dominating the entire smaller correction. Inside, write $h = n/2 + z$. Then

$$h(n-h) = n^2/4 - z^2, \quad Ch^\alpha(n-h) = C2^{-1-\alpha}n^{1+\alpha} + O_d(n^\alpha|z|).$$

Completing the square bounds the remaining maximum by $O(n^{2\alpha})$, proving the claimed upper coefficient.

For the lower bound, the standard commutator Hilbert function is

$$\dim M_\ell = (\ell-1)\binom{d+\ell-2}{d-2} \quad (\ell \ge 2).$$

With $A = (d-1)/d!$, it gives

$$h_s = As^d + O_d(s^{d-1}), \quad g_s = dAs^{d-1} + O_d(s^{d-2}), \quad h_{s+1} - h_s = g_s.$$

Choose s by $h_s \le n/2 < h_{s+1}$. Then

$$h_s = n/2 + O_d(n^\alpha), \qquad g_s = C2^{-\alpha}n^\alpha + O_d(n^{1-2/d}), \qquad r_s = O_{d,p}(\log n).$$

Apply (5). Its exponent expands as

$$(h_s + g_s - 1)(n - dr_s - h_s) = n^2/4 + \gamma_d n^{2-1/d} + O_{d,q}(n^{2-2/d} + n\log n).$$

For $d \ge 3$, $n \log n$ is absorbed by $n^{2-2/d}$. The constructed quotients are p-nilpotent for every sufficiently large integer n. Inclusion and (6) sandwich all four sequences between this lower bound and the same upper bound. □

The sharp constant uses the fraction-field rank $d-1$ of M. Its usual presentation has $\binom{d}{2}$ commutator generators; substituting that presentation rank in (7) would give a different coefficient. The entropy estimate may use any fixed presentation, because its whole contribution is below the correction scale.

5 The rank-two logarithmic term

Here $M = \mathbb{F}_q[X_1, X_2]c$, $c = [x_1, x_2]$, is cyclic. Segal's planar theorem [4, Theorem 1] gives $a_h(M) = q^{O_q(h)}$; polynomial growth in index q^h is linear-exponential growth in colength. A two-dimensional staircase also gives $g(N) \le \sqrt{2h} + 1$. Indeed, if an initial ideal has v minimal monomial generators, their ordered exponents are at least $(v-1-i, i)$, $0 \le i < v$. All monomials of total degree at most $v-2$ are standard, so $h \ge v(v-1)/2$.

Lemma 5.1 (The outer action constraint)**.** *For every finite quotient in rank two, the parameters* (2) *satisfy* $h \le p^a$.

Proof. For $Q = F_2/H$, put $C = \langle Q' \rangle_p$. The abelian restricted algebra $A = Q/C$ acts on $V = C/C^{[p]} \simeq M/N$; C acts trivially because it is abelian. This is a restricted action, and V is a cyclic $u(A)$-module generated by the commutator image. Restricted PBW gives $\dim u(A) = p^a$, with no nilpotence assumption on A. □

Proof of Theorem 1.2. The planar module count and all outer factors cost $q^{O_q(n)}$. For $h \ge 1$, the fixed-head exponent is bounded by

$$(h+g)(n-a-h) \le h(n-h) + \sqrt{2h}(n-h) - h\log_p h + O(n).$$

Put $z = h - n/2$. The tangent bound for the square root, multiplied by the nonnegative number $n-h$, gives

$$h(n-h) + \sqrt{2h}(n-h) \le n^2/4 + n^{3/2}/2 - (1 + n^{-1/2})z^2 - \sqrt{n}\, z/2.$$

Convexity gives the global tangent inequality

$$h\log_p h \ge (n/2)\log_p(n/2) + (\log_p(n/2) + 1/\ln p)z.$$

Completing the square in z leaves an $O_p(n)$ maximum, since its linear coefficient is $\sqrt{n}/2 + O_p(\log n)$. This proves the upper formula. The case $h = 0$ contributes only $q^{O_q(n)}$, and summing parameter pairs changes the logarithm by $O(\log n)$.

For the lower bound take $s = \lfloor \sqrt{n} \rfloor$. Then

$$h_s = s(s+1)/2 = n/2 + O(\sqrt{n}), \qquad g_s = s + 1 = \sqrt{n} + O(1),$$
$$r_s = \lceil \log_p(s+2) \rceil = \tfrac{1}{2} \log_p n + O_p(1).$$

The nilpotent construction (5), with $d = 2$, gives

$$(h_s + g_s - 1)(n - 2r_s - h_s) = n^2/4 + n^{3/2}/2 - \tfrac{1}{2} n \log_p n + O_q(n).$$

The rounding terms are $O(n)$: the displacement of h_s has size $O(\sqrt{n})$ and $r_s\sqrt{n} = O(n)$. Transfer to types loses only $2n$ in the logarithm, so all four counts have the same three terms. □

The boundary produces the positive $n^{3/2}$ term. The cyclic restricted action forces an outer quotient of dimension at least $\log_p h$, producing the negative $n \log_p n/2$ term. The lower family attains both mechanisms simultaneously. Substitution of $d = 2$ in Theorem 1.1 alone would miss this logarithmic scale, since it is larger than an $O(n)$ remainder.

6 Homogeneous quotients and graph applications

Proof of Theorem 1.3. Over a fixed finite extension $K/\mathbb{F}_q$, choose δ linear parameters ℓ_i so that M_K is finite over $S = K[\ell_1, \ldots, \ell_\delta]$. To see that a finite extension suffices, choose the linear forms over the algebraic closure, successively avoiding the finitely many positive-dimensional homogeneous associated primes. Their coefficients lie in a finite extension; the final finite-dimensional quotient shows finiteness over S by induction on degree. As a graded S-module M_K has rank e. A full-rank homogeneous free submodule contributes one copy of $s^{\delta-1}/(\delta-1)!$ per rank to the Hilbert leading term, and its torsion quotient has smaller dimension. Comparing with the multiplicity definition proves this assertion without a Cohen–Macaulay hypothesis.

For any $N \subset M$ of colength h, extension preserves both h and $g_R(N) = \dim N/\mathfrak{m}N$ over the respective fields. Moreover $g_R(N) \le g_S(N_K)$ since the parameter ideal lies in $\mathfrak{m}_K$. Proposition 3.1 therefore gives

$$g_R(N) \le C h^{1-1/\delta} + O_G((h+1)^{1-2/\delta}), \qquad C = \delta(e/\delta!)^{1/\delta}.$$

There is no extension-degree factor in C. Distinct submodules remain distinct after this faithfully flat extension. Applying Proposition 3.2 to a fixed free S-presentation gives an entropy bound $O_{G,q}(h^{2-2/\delta} + h\log(h+1))$; the extension degree affects only its constant.

The upper optimization in Section 4 now applies with d replaced in the exponent by δ and boundary constant C. The outer count still costs $q^{O_d(n)}$. For the lower bound the homogeneous Hilbert function gives

$$h_s = (e/\delta!)s^\delta + O_G(s^{\delta-1}), \qquad g_s = \delta(e/\delta!)s^{\delta-1} + O_G(s^{\delta-2}).$$

Choose s by $h_s \le n/2 < h_{s+1}$. Use the same ideals (4), whose outer cost is $dr_s = O_G(\log n)$. Expansion of (5) gives the stated coefficient and error $O_{G,q}(n^{2-2/\delta} + n\log n)$. The nilpotent construction and (6) prove all four models. □

6.1 Classical cut data and the growth correction

For a graph Γ let G_Γ be the ordinary free metabelian algebra with its marked edge commutators set to zero. Its restricted hull is also the quotient of F_d by the restricted ideal generated

by these edge commutators. This follows either from the universal property or by taking the abelian restricted hull of the ordinary derived relation ideal in (1).

Papadima–Suciu's cut-polynomial calculation [7, Theorems 3.4 and 4.1] gives, over an arbitrary field,

$$Q_\Gamma(z) = \sum_{j\ge 2} \sum_{|W|=j} (c(\Gamma_W) - 1) z^j, \qquad \mathrm{Hilb}(G'_\Gamma, t) = Q_\Gamma(t/(1-t)).$$

The grading here is bracket length, with commutators in degree two. All coefficients of Q_Γ are nonnegative. Its degree is δ_Γ and its leading coefficient is e_Γ; these are exactly the Krull dimension and multiplicity of the derived module. Theorem 1.3 proves Corollary 1.4. For connected noncomplete graphs, deleting a minimum vertex cut leaves a largest disconnected induced subgraph, so

$$\delta_\Gamma = d - \kappa(\Gamma), \qquad e_\Gamma = \sum_{\substack{S \text{ a minimum} \\ \text{vertex cut}}} (c(\Gamma \setminus S) - 1).$$

6.2 The path on four vertices

For P_4 the edge relations are $[x_1, x_2] = [x_2, x_3] = [x_3, x_4] = 0$. The three remaining commutators $a = [x_1, x_3]$, $b = [x_1, x_4]$, $c = [x_2, x_4]$ have the four Jacobi-relation columns

$$(X_2, 0, 0), \quad (0, X_2, -X_1), \quad (-X_4, X_3, 0), \quad (0, 0, X_3).$$

They give the complete module presentation: the free metabelian presentation has precisely the head-triple Jacobi relations, and the three edge generators have been set to zero. The relation matrix has the single free syzygy $(X_4, -X_3, X_2, -X_1)$. Its first row forces the first and third syzygy entries to be $X_4 u, X_2 u$ by coprimeness; the other two rows force the remaining entries. Thus

$$\mathrm{Hilb}(G'_{P_4}, t) = \frac{3t^2 - 4t^3 + t^4}{(1-t)^4} = \frac{t^2(3-t)}{(1-t)^3}.$$

Here $\delta = 3, e = 2$, and the four counts satisfy

$$\log_q A_n = n^2/4 + \frac{3^{2/3}}{2^{5/3}} n^{5/3} + O_q(n^{4/3}).$$

Although the source has four marked generators, its first correction is the same as in the free rank-three case.

6.3 Scope of the comparison

For a noncomplete graph, choosing a nonedge gives epimorphisms $F_d \twoheadrightarrow L_\Gamma \twoheadrightarrow F_2$. Thus its coarse quadratic growth already follows from [1], and the leading $1/4$ follows by the free-case sandwich once Theorems 1.1–1.2 are known. The graph-specific content of the transfer is its smaller correction scale and its coefficient, which the free sandwich does not determine. For a complete graph the derived module is zero, and the source is free abelian restricted; this lies outside the hypothesis $\delta \ge 2$.

The present results leave the terms inside the remainders open. They concern fixed source algebras and fixed finite fields. Uniformity in growing generator rank and the full lower-order profile require additional estimates beyond those used here.

7 Typical layers and prescribed-head laws

The growth formulas locate a quadratic maximum. Retaining the parameters of each counting stratum reveals the dimensions and power structure of the quotients that account for this maximum.

7.1 Notation and measures

Fix $q = p^f$ and generator rank $d \geq 2$. Let F_d be the free metabelian restricted Lie algebra. Use the hull decomposition in Section 2, $Z = \langle F_d' \rangle_p$ and $M = Z/Z^{[p]}$, with the usual $R = \mathbb{F}_q[X_1, \dots, X_d]$ action. A codimension-n ideal H gives

$$B = H \cap Z, \qquad b = \dim Z/B, \qquad a = n - b,$$
$$N = (B + Z^{[p]})/Z^{[p]}, \qquad h = \dim M/N, \qquad g = \dim N/\mathfrak{m}N.$$

Writing $Q = F_d/H$ and $C = \langle Q' \rangle_p$, the quantities $a = \dim Q/C$, $b = \dim C$ and $h = \dim C/C^{[p]}$ are intrinsic. The quantity g is attached to the chosen epimorphism from F_d.

The restricted-PBW argument in Section 5 has the immediate general form

$$h \leq \binom{d}{2} p^a.$$

Indeed $C/C^{[p]}$ is a restricted module for $A = Q/C$, generated by at most $\binom{d}{2}$ commutator images. Restricted PBW gives $\dim u(A) = p^a$, without a nilpotence assumption on A.

There are four uniform measures: all codimension-n ideals, those with p-nilpotent quotient, all restricted isomorphism types admitted from F_d, and the p-nilpotent types. A bad type for a marked property means one admitting at least one bad marking. Each type has at most q^{dn} kernels of epimorphisms from F_d. Thus an exceptional ideal fraction q^{-E} transfers to a type fraction at most q^{dn-E}.

7.2 Concentration of the quotient layers

Put

$$\alpha = 1 - 1/d, \quad \eta = 1 - 2/d, \quad C_d = d\big((d-1)/d!\big)^{1/d}.$$

Theorem 7.1 (Higher-rank layer concentration)**.** *For fixed $d \geq 3, q$, there are constants $K, c > 0$ such that, for each of the four measures, outside a set of probability at most $q^{-cn^{2-2/d}}$,*

$$\log_p n - K \leq a \leq Kn^{1-2/d}, \qquad |h - n/2| \leq Kn^{1-1/d}, \qquad \Big|g - C_d(n/2)^{1-1/d}\Big| \leq Kn^{1-2/d}.$$

For uniform types the last assertion can be required for every marking.

Proof. The prescribed-head bound, the fixed-module entropy bound and the outer lift bound in Sections 2–5 give a stratum exponent at most

$$(h+g)(n-a-h) + O_{d,q}(n^{2\alpha}),$$

after summing all relation modules at a fixed h; the error is uniform. Choose K_0 such that the module boundary bound is

$$g \leq B(h) := C_d h^\alpha + K_0(h+1)^\eta, \qquad \Delta = B(h) - g \geq 0.$$

Let $z = h - n/2$ and $L_d(n) = n^2/4 + C_d 2^{-1-\alpha} n^{1+\alpha}$. Outside the bulk

$$n/4 \leq h \leq 3n/4, \qquad a \leq n/8,$$

the quadratic $h(n-a-h)$ loses a fixed positive multiple of n^2. All corrections are $o(n^2)$, so this region has probability $q^{-c_0 n^2}$, using the matching lower count in Sections 2–5.

In the bulk, $n-a-h \ge n/8$ and

$$\begin{aligned}(h+g)(n-a-h) &\le h(n-h) + C_d h^\alpha (n-h) + K_0(h+1)^\eta (n-h) - ah - \Delta(n-a-h)\\ &\le L_d(n) + K_1 n^{2\alpha} - \tfrac{1}{2}z^2 - \tfrac{1}{4}na - \tfrac{1}{8}n\Delta.\end{aligned}$$

For the second line, the derivative of $C_d h^\alpha (n-h)$ is $O_d(n^\alpha)$ in the bulk; Young's inequality absorbs $O(n^\alpha |z|)$ into $z^2/2 + O(n^{2\alpha})$. Also $1+\eta = 2\alpha$.

The denominator of each measure has logarithm at least $L_d(n) - K_2 n^{2\alpha}$. The same inequalities and at most $(n+1)^2$ choices of (a,h) show, for any sufficiently large fixed u,

$$\Pr\big(\tfrac{1}{2}z^2 + \tfrac{1}{4}na + \tfrac{1}{8}n\Delta \ge un^{2\alpha},\ \text{bulk}\big) \le q^{(K_3-u)n^{2\alpha}}.$$

Increasing u gives the first two asserted scales and $\Delta = O(n^\eta)$. In the bulk the derivative of h^α is $O(n^{-1/d})$, so $g = C_d(n/2)^\alpha + O(n^\eta)$ follows. The factor q^{dn} needed to transfer bad ideals to types is absorbed into $K_3 n^{2\alpha}$. The nilpotent denominator has the same lower bound by the nilpotent construction of Sections 2–5. Finally $h \sim n/2$ and the preceding PBW inequality give the logarithmic lower bound on a, after increasing K. □

Theorem 7.2 (Rank-two near-minimal outer dimension)**.** *For fixed q, there are $K, c > 0$ such that each of the four measures assigns probability at least $1 - q^{-cn}$ to*

$$|h - n/2| \le K\sqrt{n}, \qquad |a - \log_p n| \le K, \qquad |g - \sqrt{n}| \le K.$$

For types the bound on g may again be required for every marking.

Proof. Use $g \le \sqrt{2h} + 1$, the planar relation count $q^{O_q(h)}$, and the restricted-PBW inequality $a \ge \log_p h$ from Sections 2–5. Put

$$v = a - \log_p h \ge 0, \qquad \Delta = \sqrt{2h} + 1 - g \ge 0, \qquad z = h - n/2,$$

for $h \ge 1$. The region outside the preceding bulk still loses $c_0 n^2$. In the bulk,

$$\begin{aligned}(h+g)(n-a-h) \le{}& h(n-h) + \sqrt{2h}(n-h) + (n-h)\\ &- h\log_p h - hv - \Delta(n-a-h).\end{aligned}$$

Concavity and convexity at $n/2$ give

$$\begin{aligned}h(n-h) + \sqrt{2h}(n-h) &\le n^2/4 + n^{3/2}/2 - (1+n^{-1/2})z^2 - \sqrt{n}\,z/2,\\ h\log_p h &\ge (n/2)\log_p(n/2) + (\log_p(n/2) + 1/\ln p)z.\end{aligned}$$

Completing the square and using $h \ge n/4$, $n-a-h \ge n/8$ yields

$$(h+g)(n-a-h) \le L_2(n) + K_1 n - z^2/2 - nv/4 - n\Delta/8,$$

where $L_2(n) = n^2/4 + n^{3/2}/2 - (n/2)\log_p n$. Relation modules, outer choices, lifts and the sum over strata cost $q^{O_q(n)}$. The lower count of Sections 2–5 is $q^{L_2(n) - O_q(n)}$ in each model. Thus the probability that $z^2/2 + nv/4 + n\Delta/8 > un$ is at most $q^{(K_2-u)n}$. For a sufficiently large fixed u, this gives $z = O(\sqrt{n})$, $v, \Delta = O(1)$. Consequently $a = \log_p n + O(1)$ and $g = \sqrt{n} + O(1)$. Transfer to types costs at most q^{2n} and only changes K_2. □

7.2.1 A geometric consequence for rank two

Lemma 7.3 (A staircase contains a triangular core)**.** *Let $J \subset \mathbb{F}_q[X,Y]$ be a finite-colength monomial ideal with v minimal generators and colength h. Then*

$$J \subseteq \mathfrak{m}^{v-1}, \qquad h \geq \binom{v}{2}.$$

Proof. Order minimal generator exponents as (a_i, b_i), $0 \leq i < v$, with a_i strictly decreasing and b_i strictly increasing. Then $a_i \geq v-1-i$, $b_i \geq i$. Every generator has total degree at least $v-1$, which proves both claims. □

Corollary 7.4 (Typical initial ideals have a thin extra boundary)**.** *For rank two, identify $M = Rc$. Fix a degree-compatible monomial order. Outside a set of probability at most q^{-cn} under any of the four measures, the initial ideal $J = \mathrm{in}(N)$ has*

$$v = \sqrt{n} + O_q(1), \qquad r = v-1, \qquad J \subseteq \mathfrak{m}^r, \qquad \dim \mathfrak{m}^r/J = O_q(\sqrt{n}).$$

Equivalently, its standard monomials contain the complete triangle of total degrees $< r$ and only $O_q(\sqrt{n})$ additional monomials.

Proof. The reduced Gröbner generators give $g(N) \leq v$, whereas the lemma gives $v \leq (1 + \sqrt{1+8h})/2$. The preceding theorem makes both bounds $\sqrt{n} + O(1)$. The exact sequence

$$0 \longrightarrow \mathfrak{m}^{v-1}/J \longrightarrow R/J \longrightarrow R/\mathfrak{m}^{v-1} \longrightarrow 0$$

has excess length $h - \binom{v}{2} = O(\sqrt{n})$. The triangular core describes the initial ideal; the deformation parameters of N remain free to vary within its Gröbner stratum. □

7.3 Exact nilpotent heads and the Fitting law

Let $\Omega = \mathbb{F}_q[t;\sigma]$, $\sigma(c) = c^p$. A finite-dimensional left module is a vector space with a σ-semilinear operator t. For a fixed codimension-h subspace $V \subset \mathbb{F}_q^S$, let $u_S(b,h)$ count colength-b kernels with prescribed head image V, and let $u_S^0(b,h)$ restrict to nilpotent quotients. Set

$$\phi_j(q) = \prod_{i=1}^{j}(1-q^{-i}), \quad \phi_0 = 1, \quad \phi_\infty = \prod_{i\geq 1}(1-q^{-i}), \quad \Theta_q = \phi_\infty^{-1}.$$

Proposition 7.5 (Nilpotent prescribed-head series)**.** *For $0 \leq h \leq S$,*

$$\sum_{b\geq 0} u_S^0(b,h) z^b = \frac{z^h}{\prod_{i=S-h}^{S-1}(1-q^i z)}.$$

If $h \geq 1$ and $m = b - h \geq 0$, then

$$u_S^0(h+m,h) = q^{(S-h)m} \begin{bmatrix} m+h-1 \\ h-1 \end{bmatrix}_q, \qquad q^{(S-1)m} \leq u_S^0(h+m,h) \leq \Theta_q q^{(S-1)m}.$$

For $h = 0$ the only nilpotent case has $b = 0$.

Proof. The total series for nilpotent quotients of Ω^S is $A_S^0(z) = \prod_{i=0}^{S-1}(1-q^i z)^{-1}$. To obtain it directly, project a submodule to one coordinate: its image has to be Ωt^v, and its kernel is a nilpotent-colength submodule in $S-1$ coordinates. A colength-u kernel gives q^u possible lifts of t^v, all nilpotent by extension. Thus $A_S^0(z) = (1-z)^{-1} A_{S-1}^0(qz)$.

The same subspace-lattice inversion as in Sections 2–5 applies with this series. Passing to the inverse image of a head space adds a quotient annihilated by t, so nilpotence is preserved in both directions. It gives

$$\sum_b u_S^0(b,h)z^b = z^h A_S^0(z) \prod_{i=0}^{S-h-1} (1-q^i z),$$

which cancels to the asserted expression. Expanding its h geometric factors gives the Gaussian coefficient. Normalizing its largest factor by $q^{(S-1)m}$ bounds the remaining coefficient between 1 and $\prod_{j=1}^{h-1}(1-q^{-j})^{-1} \le \Theta_q$. The argument is valid for the genuine Frobenius twist. □

Write $w_S(0) = 1$ and, for $k \ge 1$, $w_S(k) = (q^S - 1)q^{S(k-1)}$.

Proposition 7.6 (Canonical Fitting convolution)**.** *The invertible quotient kernels of Ω^S of dimension k number $w_S(k)$. For prescribed head,*

$$u_S(b,h) = \sum_{k=0}^{b-h} u_S^0(b-k,h)w_S(k).$$

Moreover, the summand is exactly the number whose t-invertible Fitting part has dimension k.

Proof. Head zero is equivalent to t invertible on a finite quotient. Putting $h = 0$ in Sections 2–5's exact series gives $\sum_k w_S(k)z^k = (1-z)/(1-q^S z)$. Every finite semilinear module has the canonical Fitting decomposition $Q = Q_0 \oplus Q_\times$. Cross homomorphisms between the nilpotent and invertible parts vanish. A tuple generates Q if and only if its projections generate both parts: a sufficiently high t-power kills Q_0 and maps the projected generating span onto $Q_\times$. The prescribed head condition concerns only Q_0. Counting epimorphisms and dividing by target automorphisms now multiplies the two kernel counts, proving the marked convolution. Equivalently the two generating functions differ by $(1-z)/(1-q^S z)$, including the endpoint $h = S$. □

Corollary 7.7 (Uniform and asymptotically exact nilpotent proportion)**.** *For $1 \le h \le S$ and $m = b - h$,*

$$\frac{u_S^0(b,h)}{u_S(b,h)} = q^{-m+O_q(1)}$$

uniformly in all parameters. As $h, m \to \infty$, uniformly for $S \ge h$,

$$\frac{u_S^0(h+m,h)}{u_S(h+m,h)} = q^{-m}\big(1 + O_q(q^{-h} + q^{-m})\big).$$

Proof. Combine Proposition 7.5 with the uniform head bounds of Proposition 2.1 for the first assertion. Put

$$A_h(\ell) = q^{-(h-1)\ell} \begin{bmatrix} \ell + h - 1 \\ h - 1 \end{bmatrix}_q = \frac{\prod_{i=1}^{\ell}(1 - q^{-(h-1+i)})}{\phi_\ell}.$$

Uniformly in ℓ, $A_h(\ell) = \phi_\ell^{-1}(1 + O_q(q^{-h}))$. The Fitting convolution gives

$$V_{S,h,m} := q^{-Sm}u_S(h+m,h) = (1-q^{-S}) \sum_{\ell=0}^{m-1} q^{-\ell}A_h(\ell) + q^{-m}A_h(m).$$

Euler's identity $\sum_{\ell \ge 0} q^{-\ell}/\phi_\ell = \Theta_q$ implies $V_{S,h,m} = \Theta_q + O_q(q^{-h} + q^{-m})$. Also $A_h(m) = \Theta_q + O_q(q^{-h} + q^{-m})$. Divide $q^{(S-1)m}A_h(m)$ by $q^{Sm}V_{S,h,m}$. □

Theorem 7.8 (Universal excess and partition laws)**.** *Choose a kernel uniformly with fixed (S, b, h, V) and $h \geq 1$. Let $k = \dim Q_\times$ and $\ell = \dim Q_0 - h = m - k$. As $h, m \to \infty$, uniformly for $S \geq h$,*

$$\Pr(\ell = j) \longrightarrow \phi_\infty \frac{q^{-j}}{\phi_j}, \qquad j \geq 0,$$

in total variation, with error $O_q(q^{-h} + q^{-m})$. In particular ℓ is tight with a uniform bound $\Pr(\ell \geq L) \leq C_q q^{-L}$.

If the nilpotent Jordan partition of Q_0 is λ with h parts, delete its first column, obtaining $\nu = (\lambda_1 - 1, \ldots, \lambda_h - 1)$ with zero parts omitted. Then its limiting distribution in total variation is

$$\Pr(\nu) \longrightarrow \frac{\phi_\infty}{a_q(\nu)}, \qquad a_q(\nu) = q^{\sum_j (\nu'_j)^2} \prod_{r \geq 1} \phi_{m_r(\nu)}.$$

This is the classical Hall/Cohen–Lenstra partition measure [8].

Proof. For $0 \leq j < m$ the exact excess mass is

$$\frac{(1 - q^{-S}) q^{-j} A_h(j)}{V_{S,h,m}},$$

and its endpoint mass at $j = m$ is $q^{-m} A_h(m)/V_{S,h,m}$. The uniform bounds and the preceding normalizer estimate give both the tail bound and the total variation estimate by summing a geometric series.

For a partition ν of size j and length $r \leq h$, its associated λ has first column h. A nilpotent semilinear module has one chain type for each partition, with $|\operatorname{Aut} Q_\lambda| = q^{\sum_i (\lambda'_i)^2} \prod_t \phi_{m_t(\lambda)}$. This follows by counting chain homomorphisms and units on the multiplicity spaces; Frobenius is bijective and does not alter the count. A quotient of nilpotent dimension $h + j$ and this type with prescribed head has exactly

$$q^{Sj} \frac{\phi_h}{\phi_{h-r}} \frac{1}{a_q(\nu)}$$

kernels: count generator images inducing a head isomorphism, divide by automorphisms, and use $\sum_i (\lambda'_i)^2 = h^2 + \sum_i (\nu'_i)^2$. Multiplication by $w_S(m - j)$ and division by $u_S(h + m, h)$ gives

$$\Pr(\nu) = \frac{\kappa_{S,m-j}}{V_{S,h,m}} \frac{\phi_h}{\phi_{h-r}} \frac{1}{a_q(\nu)}, \quad \kappa_{S,k} = \begin{cases} 1 - q^{-S}, & k > 0, \\ 1, & k = 0. \end{cases}$$

For each fixed partition this tends to the displayed limit. The uniform excess tail makes these pointwise limits tight, yielding total variation convergence and normalization. This also proves $\sum_{\nu \vdash j} a_q(\nu)^{-1} = q^{-j}/\phi_j$. The limiting inverse-automorphism measure is classical; see Fulman [8], Sections 1–2. □

8 Outer compatibility and global partition laws

The prescribed-head counts become global laws once the actual outer compatibility equations are retained. Their contribution is an exact nonnegative mixture over a bounded range of head ranks. Uniformity in that rank then preserves the partition laws.

8.1 The exact conditional outer count

Let $L = L_G$ be a hull in Sections 2–5, including F_d. Write $Z = \langle G' \rangle_p$, $M = G'$, $A = L/Z \simeq \Omega^d$, where $\Omega = \mathbb{F}_q[t;\sigma]$ and $\sigma(c) = c^p$. Choose the source generators $x_1, \dots, x_d$. Fix an R-submodule $N \subset M$ of colength h, where $R = \mathbb{F}_q[X_1, \dots, X_d]$. Set

$$C_N = \langle \mathfrak{m}N \rangle_p, \quad U = M/\mathfrak{m}N, \quad V = N/\mathfrak{m}N, \quad W = M/N, \quad S = \dim U = h+g, \quad g = \dim V.$$

The quotient $\Gamma = Z/C_N$ is the free abelian restricted algebra on U. The ideal C_N is stable under L: its ordinary head is the R-submodule $\mathfrak{m}N$, and its positive powers are central.

Fix a finite-colength Ω-submodule $O \subset A$, of colength a. It is free of rank d. Choose an Ω-basis $o_1, \dots, o_d$ and lifts y_j in the head-tower vector complement of Z. Thus $[x_i, y_j] \in M$. Define

$$T_i : W \longrightarrow U, \quad T_i(\bar{u}) = [x_i, u] \bmod \mathfrak{m}N, \qquad c_{ij} = [x_i, y_j] \bmod \mathfrak{m}N.$$

The map T_i is well defined because $[x_i, N] \subset \mathfrak{m}N$. For $\alpha = (\alpha_1, \dots, \alpha_d) \in W^d$, put

$$D(\alpha) = \operatorname{span}_{\mathbb{F}_q}\{c_{ij} + T_i\alpha_j : 1 \le i, j \le d\}.$$

Let $c_r(N, O)$ count the tuples with $D(\alpha) \subset V$ and $\dim D(\alpha) = r$. These nonnegative integers depend on the chosen basis and lifts only through the equivalent counting problem; their individual presentation is not claimed canonical. Only $0 \le r \le \min(g, d^2)$ can occur.

Theorem 8.1 (A bounded-rank mixture)**.** *Let $K_{N,O}(b)$ count restricted ideals $H \subset L$ such that*

$$(H + Z)/Z = O, \qquad \dim Z/(H \cap Z) = b, \qquad (H \cap Z + Z^{[p]})/Z^{[p]} = N.$$

For $m = b - h \ge 0$,

$$\boxed{K_{N,O}(b) = q^{dm} \sum_{r=0}^{\min(g,d^2)} c_r(N, O)\, u_{S-r}(b, h).}$$

Replacing u by u^0 counts those with nilpotent inner module $Z/(H \cap Z)$. If A/O is p-nilpotent, this also counts precisely the p-nilpotent whole quotients. If A/O is not p-nilpotent, there are no p-nilpotent whole quotients in this stratum. The formula can be refined by every Fitting or nilpotent block statistic by using its corresponding prescribed-head count in each summand.

Proof. Write $B = H \cap Z$. Any such B contains C_N. Conversely a restricted submodule $B/C_N \subset \Gamma$ with prescribed head V is a Lie ideal in L/C_N: its ordinary head belongs to V, which is killed by every X_i, and every positive restricted power in Z is central. Let $C = Z/B$. Then $C^{[p]}$ is central and $C/C^{[p]} \simeq W$.

For any ideal H with inner kernel B, one has $[H, L] \subset B$, since the derived ideal is contained in Z. Thus H/B is central in L/B and projects isomorphically onto O. It is specified uniquely by the lifts $y_j + z_j$ of its fixed free Ω-basis, where $z_j \in C$. Conversely, arbitrary central lifts of this basis extend uniquely to a restricted-linear map $O \to Z(L/B)$; its projection to O is the identity. Its image gives the desired ideal and has zero intersection with C. No extra relation needs checking because O is free over Ω and the chosen lifts are central.

Fix the heads $\alpha_j = z_j + C^{[p]} \in W$. There are exactly q^{dm} choices of the z_j with these heads. They have identical commutators with all x_i, because $C^{[p]}$ is central. Centrality is equivalent to

$$c_{ij} + T_i\alpha_j \in B/C_N \quad \text{for every } i, j.$$

Checking the source generators suffices: brackets and restricted powers then show commutation with the entire restricted algebra they generate. The displayed condition is $\Omega D(\alpha) \subset B/C_N$. It is impossible unless $D(\alpha) \subset V$.

For a feasible tuple of rank r, quotienting Γ by ΩD leaves a free abelian restricted algebra of rank $S-r$, and its prescribed head V/D still has codimension h. The quotient dimension and its Fitting data are unchanged. Its kernel count is exactly $u_{S-r}(b,h)$. Multiplying by q^{dm} and summing all feasible tuples proves the formula with no overcount: for a fixed ideal each outer basis vector has exactly one lift modulo B in H/B.

Finally L/H is an extension of A/O by the abelian restricted ideal C. Such an extension is p-nilpotent if and only if both pieces are: an iterated p-power first takes an element into C, and further powers kill it. This proves the last assertions. □

Computability. Feasibility of α is the affine linear system $\bar{c}_{ij} + \bar{T}_i\alpha_j = 0$ in W. After solving it, the only remaining classification is the rank of a matrix with at most d^2 columns in V. The theorem does not require explicitly evaluating the rank-stratum cardinalities in order to transfer a uniform limiting law.

8.2 Uniform prescribed-head inputs

Write $\phi_j = \prod_{i=1}^{j}(1-q^{-i})$, $\phi_0 = 1$, $\phi_\infty = \prod_{i\ge 1}(1-q^{-i})$ and $\Theta_q = \phi_\infty^{-1}$. Section 7 proves, for $s \ge h \ge 1$ and $m = b - h$,

$$u_s^0(h+m,h) = q^{(s-1)m}A_h(m),$$
$$A_h(m) = \frac{\prod_{i=1}^{m}(1-q^{-(h-1+i)})}{\phi_m},$$
$$\frac{u_s^0(h+m,h)}{u_s(h+m,h)} = q^{-m}\big(1+O_q(q^{-h}+q^{-m})\big).$$

The relative error is uniform in $s \ge h$. The first identity follows from the nilpotent head generating series $z^h\prod_{i=s-h}^{s-1}(1-q^iz)^{-1}$; the last follows by separating the invertible Fitting factor $(1-z)/(1-q^sz)$.

For the inner quotient, let Q_0 be its nilpotent Fitting part and λ its Jordan partition, with h parts. Delete the first column, obtaining ν with $|\nu| = \dim Q_0 - h$. Define

$$a_q(\nu) = q^{\sum_j(\nu_j')^2}\prod_t \phi_{m_t(\nu)}, \qquad \mu_q(\nu) = \frac{\phi_\infty}{a_q(\nu)}.$$

The prescribed-head law converges to μ_q uniformly for $s \ge h$ as $h, m \to \infty$. More precisely its total variation error is

$$O_q(q^{-h/2}+q^{-m/2}).$$

Indeed the size law has error $O_q(q^{-h}+q^{-m})$ and tail $O_q(q^{-L})$. On $|\nu| \le \min(h,m)/2$, its exact individual mass is

$$\frac{\kappa_{s,m-|\nu|}}{V_{s,h,m}}\frac{\phi_h}{\phi_{h-\ell(\nu)}}\frac{1}{a_q(\nu)}, \quad V_{s,h,m} = q^{-sm}u_s(h+m,h),$$

where $\kappa_{s,k} = 1-q^{-s}$ for $k>0$ and 1 for $k=0$. Here $V_{s,h,m} = \Theta_q + O_q(q^{-h}+q^{-m})$. The displayed range controls the remaining ratio uniformly; the complementary mass is geometrically small. The measure μ_q and its Euler/Hall normalization are classical, not new probability measures; see Fulman [8], Sections 1–2.

Corollary 8.2 (Compatibility preserves the conditional laws)**.** *For every nonempty (N,O,b) stratum, the inner Fitting-partition law has the same uniform limit μ_q. Its inner-nilpotence probability is*

$$q^{-m}\big(1+O_q(q^{-h}+q^{-m})\big).$$

These statements are uniform over the source-dependent compatibility coefficients c_r, provided $h, m \to \infty$.

Proof. The common lift factor q^{dm} cancels in probabilities. Each summand is the prescribed-head model of rank $s = S - r \geq h$. The conditional law is a convex mixture of those models, so its error is no larger than the uniform bound for its components. The relative nilpotence estimate is uniform as well, and survives the same weighted averaging. □

8.3 Global laws under uniform ideal counting

Assume the source is F_d with $d \geq 2$, or one of the fixed homogeneous hulls L_G of Sections 2–5 with derived-module dimension $\delta \geq 2$. The field and source are fixed. Let H be uniform among all its codimension-n restricted ideals, and set

$$Q = L/H, \quad C = \langle Q' \rangle_p, \quad a = \dim Q/C, \quad b = \dim C, \quad h = \dim C/C^{[p]}, \quad m = b - h.$$

The Fitting decomposition and the partition ν below refer to the semilinear restricted-power operator on the abelian ideal C. Its nilpotent part need not be a Lie ideal of the whole quotient. Its invertible part is always a central restricted ideal.

A cautionary four-dimensional example. Over $\mathbb{F}_2$, take a basis x, y, u, v, with the only nonzero basic brackets $[x, y] = [x, u] = u + v$, and prescribe

$$x^{[2]} = x, \quad y^{[2]} = u^{[2]} = 0, \quad v^{[2]} = v.$$

Jacobi holds, and $\mathrm{ad}(z)^2 = \mathrm{ad}(z^{[2]})$ on the basis; the usual restricted sum formula therefore defines a restricted algebra. It is generated by x, y, since $(u+v)^{[2]} = v$. Here $C = \mathrm{span}\{u, v\}$, $C_0 = \mathrm{span}\{u\}$ and $C_\times = \mathrm{span}\{v\}$. Thus C_0 is not an ideal, as $[x, u] = u + v$. The Fitting statements in this note describe the inner restricted-power operator, and do not assert a product decomposition of the whole Lie algebra.

Lemma 8.3 (Macroscopic concentration)**.** *For every fixed $\epsilon > 0$, outside an ideal fraction $q^{-c_\epsilon n^2}$,*

$$a \leq \epsilon n, \qquad |h - n/2| \leq \epsilon n.$$

The same conclusion holds when sampling only p-nilpotent quotients.

Proof. The stratum exponent is at most $h(n - a - h) + o(n^2)$ uniformly in its parameters. Here the boundary contribution is $o(n^2)$, relation-module entropy is $o(n^2)$, and outer choices cost $O(n)$ in the logarithm, by Sections 2–5. The lower construction gives $n^2/4 + o(n^2)$ even for p-nilpotent quotients. If $a > \epsilon n$, the quadratic maximum is at most $(1 - \epsilon)^2 n^2/4$; if $|h - n/2| > \epsilon n$, it is at most $n^2/4 - \epsilon^2 n^2$. The polynomial number of parameter pairs does not change the exponential gap. □

Theorem 8.4 (Global Hall law and a large central invertible part)**.** *Under uniform counting of all codimension-n ideals,*

$$\|\mathcal{L}(\nu) - \mu_q\|_{\mathrm{TV}} = O_{L,q}(q^{-\kappa n})$$

for some $\kappa > 0$. Consequently

$$\dim C_\times = b - h - O_{\Pr}(1), \qquad \frac{\dim C_\times}{n} \longrightarrow \frac{1}{2}$$

in probability. The subspace $C_\times$ is a central restricted ideal.

Proof. On the bulk $a \leq n/8$, $n/4 \leq h \leq 3n/4$, both $h, m \geq n/8$. Lemma 8.3 makes its complement q^{-cn^2}. Condition on the actual N, O, b and apply Corollary 8.2; its error is uniformly $O_q(q^{-n/16})$. Mixing the strata proves the total variation statement. The size $|\nu|$ is tight by the geometric tail of μ_q, and $\dim C_\times = b - h - |\nu|$. The macroscopic concentration yields the normalized dimension limit. Finally t is invertible on $C_\times$, so $C_\times \subset C^{[p]} \subset Z(Q)$. □

Theorem 8.5 (Global rarity of inner nilpotence)**.** *For uniform all-ideal counting,*

$$\Pr(C \text{ is } p\text{-nilpotent}) = q^{-n/2+o(n)}.$$

In particular

$$\Pr(Q \text{ is } p\text{-nilpotent}) \le q^{-n/2+o(n)},$$

so whole p-nilpotence has probability tending to zero. A matching lower bound for whole p-nilpotence is not established here.

Proof. Use the uniform finite bounds $c_q q^{-m} \le \Pr(C \text{ nil} \mid N, O, b) \le C_q q^{-m}$ whenever $h \ge 1$. On the ϵ-bulk of Lemma 8.3,

$$(1/2 - 2\epsilon)n \le m \le (1/2 + \epsilon)n.$$

The exceptional probability is $q^{-c_\epsilon n^2}$. Averaging gives

$$c_q(1 - q^{-c_\epsilon n^2})q^{-(1/2+\epsilon)n} \le \Pr(C \text{ nil}) \le C_q q^{-(1/2-2\epsilon)n} + q^{-c_\epsilon n^2}.$$

Take logarithms, divide by n, and then let $\epsilon \downarrow 0$. Whole nilpotence implies inner nilpotence, proving the inequality. This argument uses uniform relative estimates; total variation convergence alone would not determine this rare-event exponent. □

8.4 The global nilpotent ensemble has one giant chain

Now sample H uniformly only among those with $Q = L/H$ p-nilpotent. The operator on C is nilpotent. Write its partition as λ with h parts, and put $\nu = \lambda - (1^h)$, so $|\nu| = m = b - h$. Delete the largest part of ν, obtaining τ. When $m = 0$ define $\tau = \varnothing$; this event leaves the bulk as $n \to \infty$.

Lemma 8.6 (An exact conditional nilpotent law)**.** *In every nonempty nilpotent (N, O, b) stratum, for $\nu \vdash m$ with $r = \ell(\nu) \le h$,*

$$\Pr(\nu \mid N, O, b, Q \text{ nil}) = \frac{q^m}{A_h(m)} \frac{\phi_h}{\phi_{h-r}} \frac{1}{a_q(\nu)}.$$

This is independent of every compatibility coefficient c_j.

Proof. For a prescribed-head model of any rank $s \ge h$, the partition count and the total nilpotent count are respectively

$$\frac{q^{sm}\phi_h}{\phi_{h-r}a_q(\nu)}, \qquad q^{(s-1)m}A_h(m).$$

Their ratio is independent of s. The rank mixture of Theorem 8.1 therefore has exactly the same ratio, not just an asymptotically matching one. □

Let $\rho_m(\nu) = \phi_m q^m / a_q(\nu)$ on partitions of m. This is the classical inverse-automorphism measure conditioned on size. Uniformly in m, the law in Lemma 8.6 differs from ρ_m by $O_q(q^{-h})$ in total variation. Here are details. For $m \ge 1$ and $1 \le r \le m$, the head identity implies

$$\sum_{\substack{|\nu|=m \\ \ell(\nu)=r}} \frac{1}{a_q(\nu)} = \frac{q^{-rm}}{\phi_r} \begin{bmatrix} m-1 \\ r-1 \end{bmatrix}_q.$$

The Gaussian coefficient bound makes $\rho_m(\ell(\nu) = r) \le C_q q^{-r(r-1)}$, uniformly in m. The factor ϕ_h/ϕ_{h-r} differs from 1 by at most $C_q q^{-h+r}$ when $r \le h$. Its expectation under ρ_m is $O_q(q^{-h})$, and the omitted tail $r > h$ is smaller still. Normalization gives the claimed distance.

Lemma 8.7 (Condensation of the conditioned partition)**.** *For $\nu \sim \rho_m$, remove the largest row and call the remainder τ. As $m \to \infty$ its law tends in total variation, with error $O_q(q^{-m})$, to*

$$\mu_q^{(1/q)}(\tau) = \frac{\phi_\infty}{1-q^{-1}} \frac{q^{-|\tau|}}{a_q(\tau)}.$$

In particular $m - \nu_1 = |\tau| = O_{\Pr}(1)$.

Proof. If $k = |\tau| < m/2$, the largest row has length $m-k$ and is strictly larger than every other row. The automorphism formula gives

$$a_q((m-k,\tau)) = q^{m+k}(1-q^{-1})a_q(\tau).$$

Hence the exact probability of this remainder is $\phi_m q^{-k}/((1-q^{-1})a_q(\tau))$. Summing over partitions of k gives exact mass $\phi_m q^{-2k}/((1-q^{-1})\phi_k)$ for $k < m/2$. Euler's identity says

$$\sum_{k\geq 0} \frac{q^{-2k}}{\phi_k} = \frac{1-q^{-1}}{\phi_\infty}.$$

For every integer $K < m/2$, the probability of $k \geq K$ is at most $C_q q^{-2K}$, since $\phi_m \geq \phi_\infty$ and the omitted limit mass is a geometric tail. Choosing K near $m/2$ bounds the noncondensed part by $O_q(q^{-m})$. On its complement $\phi_m/\phi_\infty = 1 + O_q(q^{-m})$, proving total variation convergence. The limiting measure is the classical Hall/Cohen–Lenstra family with parameter $u = 1/q$, not a new definition. □

Theorem 8.8 (Giant-chain law for all nilpotent ideals)**.** *Under uniform counting of all codimension-n ideals with p-nilpotent quotient, τ converges in total variation to $\mu_q^{(1/q)}$, with error $O_{L,q}(q^{-\kappa n})$ for some $\kappa > 0$. The inner restricted-power operator has one chain of length*

$$m + 1 - O_{\Pr}(1) = n/2 + o_{\Pr}(n),$$

all other chains have bounded length in probability, and the excess size in these other chains is tight. Moreover

$$\Pr\big(\lambda = (m+1, 1^{h-1})\big) \longrightarrow \frac{\phi_\infty}{1-q^{-1}} = \prod_{j\geq 2}(1-q^{-j}).$$

Proof. On the macroscopic bulk, $h, m \geq n/8$. Combine the exact conditional law, its $O(q^{-h})$ comparison with ρ_m, and Lemma 8.7, then mix all (N, O, b) strata. The complementary probability is q^{-cn^2} by Lemma 8.3. Reinstating the deleted first column gives

$$\lambda = (m - |\tau| + 1, \tau_1 + 1, \ldots, \tau_r + 1, 1^{h-r-1})$$

on the condensed event. The hook is exactly $\tau = \varnothing$. Its stated limit can also be checked directly: for $m \geq 1$ its conditional probability is $(1-q^{-h})/((1-q^{-1})A_h(m))$. □

Corollary 8.9 (Height of the whole nilpotent quotient)**.** *Let $e_p(Q) = \min\{e : x^{[p^e]} = 0$ for every $x \in Q\}$. In the nilpotent ideal ensemble,*

$$\frac{e_p(Q)}{n} \longrightarrow \frac{1}{2}$$

in probability. This is restricted-power height, not lower-central nilpotency class.

Proof. The height of C is λ_1, while that of the abelian outer quotient is at most its dimension a. Iterating powers first into C and then to zero gives $\lambda_1 \leq e_p(Q) \leq a + \lambda_1$. Use $a = o_{\Pr}(n)$ and $\lambda_1/n \to 1/2$. □

9 Central invertible factors and adjacent growth

Ordinary Lie nilpotence and nilpotence of the restricted power impose different conditions. Separating them produces a canonical central factor, an exact convolution, and a first-difference law for ideal growth. The central-torus background is classical; see [11].

9.1 Two different forms of nilpotence

Use the source $L = L_G$ and notation of the preceding notes, with $M = G'$, $Z = \langle M \rangle_p$, $L/Z \simeq \Omega^d$, $\Omega = \mathbb{F}_q[t;\sigma]$, $\sigma(c) = c^p$. For $Q = L/H$ put

$$C = \langle Q' \rangle_p, \quad A_Q = Q/C, \quad N = (H \cap Z + Z^{[p]})/Z^{[p]}, \quad W = M/N \simeq C/C^{[p]}.$$

An $\mathfrak{m}$-primary relation module here means $\mathfrak{m}^s M \subseteq N$ for some s, including the case $W = 0$. It refers to support only at the origin, not to a choice of homogeneous representatives.

Lemma 9.1 (An intrinsic locality condition)**.** *The ordinary Lie algebra underlying Q is nilpotent if and only if N is $\mathfrak{m}$-primary. This does not require C to be p-nilpotent.*

Proof. If $\mathfrak{m}^s M \subset N$, the image of $\mathfrak{m}^s M$ lies in $C^{[p]} \subset Z(Q)$. Every source-generator adjoint on M is X_i, and every positive restricted tower acts by a positive power of an X_i. Consequently successive brackets eventually enter this central image, and the next bracket is zero. This proves ordinary nilpotence. Conversely ordinary nilpotence makes all X_i nilpotent on W. They commute, so a sufficiently large power of $(X_1, \dots, X_d)$ kills W. □

Theorem 9.2 (Canonical splitting under both hypotheses)**.** *Suppose Q is ordinary Lie nilpotent and C is p-nilpotent. There is a canonical decomposition as restricted Lie algebras*

$$Q = Q_{\mathrm{nil}} \oplus T,$$

where Q_{nil} is p-nilpotent and T is central abelian with invertible restricted-power operator. More precisely, Q_{nil} is the set of all p-nilpotent elements, and T is the invertible Fitting part of the center $Z(Q)$. The projection $T \to (A_Q)_\times$ is an isomorphism, and Q_{nil} is the inverse image of $(A_Q)_0$.

Proof. The inverse image U of the nilpotent Fitting part of A_Q is a restricted ideal. Every element first powers into C and then to zero, so U is p-nilpotent. Conversely every p-nilpotent element belongs to U.

Ordinary nilpotence implies that a uniformly sufficiently high restricted power of every $x \in Q$ is central, since $\mathrm{ad}(x^{[p^j]}) = (\mathrm{ad}\, x)^{p^j}$. For any $v \in (A_Q)_\times$, lift v to x. Such high powers of the lifts show that the image of $Z(Q)$ covers $(A_Q)_\times$: the restricted-power operator is surjective on that Fitting part. Apply the ordinary semilinear Fitting decomposition to the abelian restricted algebra $Z(Q)$. Its invertible part T maps onto $(A_Q)_\times$, since its nilpotent part can only map into $(A_Q)_0$. The kernel is contained in the p-nilpotent C, hence is zero. Thus $Q = U \oplus T$, and the summands commute because T is central. Both constructions are intrinsic, proving canonicity. □

Why the inner hypothesis matters. It is not enough that the ordinary Lie algebra be nilpotent. Over $\mathbb{F}_2$, let $[x, y] = z$ be Heisenberg and prescribe $x^{[2]} = y^{[2]} = 0$, $z^{[2]} = z$. Then x, y are p-nilpotent but $x + y$ is not: $(x + y)^{[2]} = z$. Here $C = \mathbb{F}_2 z$ is invertible, and the set of p-nilpotent elements is not a subspace. The hypothesis on ordinary nilpotence is also necessary: the two-dimensional algebra $[x, y] = y$, $x^{[2]} = x$, $y^{[2]} = 0$ has p-nilpotent C, but its toral direction is not central.

9.2 An exact marked convolution, and a type identity

Let P_n count all codimension-n ideals with p-nilpotent quotient, and let D_n count those with both ordinary-nilpotent quotient and p-nilpotent C. Put $P_0 = D_0 = 1$ and

$$w_d(0) = 1, \qquad w_d(k) = (q^d - 1)q^{d(k-1)} \quad (k \geq 1).$$

This is the number of k-dimensional invertible abelian quotient kernels of Ω^d, as in Section 7.

Theorem 9.3 (Central torus convolution)**.** *Exactly, for every $n \geq 0$,*

$$\boxed{D_n = \sum_{k=0}^{n} P_{n-k} w_d(k).}$$

The summand counts precisely those with $\dim T = k$. Equivalently, as formal series (no analytic convergence is asserted),

$$D(z) = P(z)\frac{1-z}{1-q^d z}.$$

The identity also holds with a fixed ordinary relation module N in both counts, with zero terms where no such quotient exists.

Proof. Choose an ideal H_0 with p-nilpotent quotient Q_0, and an invertible abelian quotient kernel J of $L/Z \simeq \Omega^d$. The combined map

$$L \longrightarrow Q_0 \oplus (\Omega^d/J)$$

is surjective. In fact its two projections are onto. A high restricted power kills all of Q_0 and is invertible on the second factor, so the image contains the whole second factor; subtracting then gives the first. This argument works because the second factor is central in the direct sum. The product kernel is $H_0 \cap \pi^{-1}J$, where $\pi : L \to L/Z$. It has codimension $\dim Q_0 + \dim \Omega^d/J$.

Conversely Theorem 9.2 recovers these two kernels uniquely from the product quotient. Every homomorphism from a p-nilpotent algebra to an invertible abelian algebra is zero, and every homomorphism in the reverse direction is zero as well. Thus there is no ambiguity from mixing the two factors. Finally M maps only to Q_0, so the relation module N is unchanged. □

Corollary 9.4 (An exact identity for unmarked types)**.** *Let $P_n^{\mathrm{type}}, D_n^{\mathrm{type}}$ be the corresponding numbers of unmarked restricted isomorphism types admitted from this fixed L. Let $v_d(k)$ be the number of k-dimensional invertible abelian restricted isomorphism types admitted from Ω^d. Then*

$$D_n^{\mathrm{type}} = \sum_{k=0}^{n} P_{n-k}^{\mathrm{type}} v_d(k).$$

Proof. The two factors are canonical and are preserved by every restricted isomorphism. Conversely every admitted pair gives an admitted product by the same surjectivity argument. The required number of source generators does not add: the same d source elements can project onto generating tuples in both factors. We do not substitute $w_d(k)$ for $v_d(k)$; these count different objects. □

9.3 Adjacent-dimension smoothness

Now assume the source is F_d, $d \geq 2$, or a homogeneous hull of Section 2.1 with derived-module dimension $\delta \geq 2$. The field and source are fixed. Let C_n count all codimension-n ideals.

Theorem 9.5 (The first difference of logarithmic growth)**.** *For $E_n = C_n$ or P_n,*

$$\boxed{\log_q \frac{E_n}{E_{n-1}} = \frac{n}{2} + o(n).}$$

For the free source F_d the remainder can be sharpened to $O_{d,q}(n^{1-1/d})$, including $O_q(\sqrt{n})$ when $d = 2$. This does not follow merely by subtracting the known asymptotic formulas, whose remainders are too large for that operation.

Proof. Use the exact compatibility strata (N, O, b) in Section 8. Keep N, O fixed and replace b by $b + 1$, so total codimension increases by one. Write $h = \dim M/N$, $S = h + g(N)$ and $m = b - h$. For $h \geq 1$ the uniform prescribed-head bounds imply, for absolute positive constants depending only on q,

$$c_q q^s \leq \frac{u_s(h+m+1,h)}{u_s(h+m,h)} \leq C_q q^s, \qquad c_q q^{s-1} \leq \frac{u_s^0(h+m+1,h)}{u_s^0(h+m,h)} \leq C_q q^{s-1}.$$

Every nonempty compatibility stratum is a nonnegative mixture of these models, with $h \leq s = S - r \leq S$, and its lift factor is q^{dm}. Hence, with $\epsilon = 0$ for all ideals and 1 for nilpotent ideals,

$$c_q q^{d+h-\epsilon} \leq \frac{K^{\epsilon}_{N,O}(b+1)}{K^{\epsilon}_{N,O}(b)} \leq C_q q^{d+h+g-\epsilon}.$$

For the nilpotent sequence the allowed outer kernels O must be nilpotent; that condition is unchanged by increasing b.

Both ensembles have a macro-bulk of probability $1-o(1)$ with $h = n/2+o(n)$, $b-h \geq n/8$; the module boundary gives $g = o(n)$ uniformly there. For a fully uniform use of the $o(n)$ notation, first fix any positive macroscopic tolerance, apply the bulk theorem, and then let the tolerance decrease after taking limits. Increase b in the $(n-1)$-bulk. The strata remain distinct and nonempty, giving $E_n/E_{n-1} \geq q^{n/2-o(n)}(1-o(1))$. Decrease b in the n-bulk. Its inner excess is positive, so all the resulting strata are nonempty and distinct. The upper ratio bound gives $(1-o(1))E_n \leq q^{n/2+o(n)}E_{n-1}$. These two inequalities prove the result.

For F_d, use Section 7's quantitative bulk $h = n/2 + O(n^{1-1/d})$; the uniform boundary also bounds $g = O(n^{1-1/d})$. Its exceptional probability tends to zero. The same two comparisons give the asserted quantitative remainder. □

9.4 The outer central torus has an exponentially small cost

Theorem 9.6 (Dominance of the zero toral factor)**.** *In uniform sampling from the D_n ideals,*

$$\Pr(Q \text{ is } p\text{-nilpotent}) = \frac{P_n}{D_n} = 1 - q^{-n/2+o(n)}.$$

More precisely,

$$D_n - P_n = (q^d - 1)P_{n-1}\big(1 + q^{-n/2+o(n)}\big).$$

Conditionally on a nonzero toral factor, its dimension is one with probability $1 - q^{-n/2+o(n)}$. For every fixed $r \geq 1$,

$$\Pr(\dim T \geq r) = q^{-rn/2+o(n)}.$$

For free sources the exponents may use the remainder $O_{d,q,r}(n^{1-1/d})$ in place of $o(n)$.

Proof. By Theorem 9.5, for each fixed k, $P_{n-k}/P_n = q^{-kn/2+o(n)}$. In the exact convolution let $a_k = w_d(k)P_{n-k}$. For $1 \le k < n/2$, its consecutive ratio satisfies uniformly

$$\frac{a_{k+1}}{a_k} = q^d \frac{P_{n-k-1}}{P_{n-k}} \le q^{-n/5}$$

for all sufficiently large n. Thus the sum on this range is dominated by its first term. For $k \ge n/2$, the growth inputs $\log_q P_j = j^2/4 + o(j^2)$, uniformly in the bounded initial range as needed, give a total contribution at most $q^{-cn^2}P_n$ for some $c > 0$, since $w_d(k) \le q^{dk}$. To obtain the stated sharper relative error after the first term, apply the same argument beginning at a_2: $\sum_{k\ge 2} a_k = a_2(1 + o(1))$ and $a_2/a_1 = q^{-n/2+o(n)}$. The proof beginning at any fixed r is identical. Division by $D_n = P_n(1 + q^{-n/2+o(n)})$ gives all probability claims. □

9.5 Factoring the whole-quotient probability

Let B_n count ideals whose inner quotient C is p-nilpotent, and retain $C_n = c_n$, $P_n = c_n^{\mathrm{nil}}$ and D_n from above. Put

$$\pi_n = D_n/B_n = \Pr(Q \text{ is ordinary Lie nilpotent} \mid C \text{ is } p\text{-nilpotent}).$$

Exactly,

$$\frac{P_n}{C_n} = \frac{B_n}{C_n}\,\pi_n\,\frac{P_n}{D_n}.$$

The first factor has logarithmic rate $-1/2$, and the last tends to one with the rate established above. Section 10 determines the remaining linear exponent in rank two. For higher rank it isolates the cost of forcing the ordinary relation module to be supported at the origin.

10 The rank-two whole-quotient nilpotence rate

In two variables, near-extremal relation ideals have only sublinear counting entropy. A comparison that preserves their colength and generator number then controls the cost of origin support. Together these give the whole-quotient exponent 1/2.

10.1 Statement and notation

Fix $q = p^f$ and the free metabelian restricted source F_2 on x, y. Its ordinary commutator module is $M = Rc$, $R = \mathbb{F}_q[X, Y]$, $c = [x, y]$. Use $\mathfrak{m} = (X, Y)$, $Z = \langle M \rangle_p$, and $F_2/Z = \Omega^2$, $\Omega = \mathbb{F}_q[t; \sigma]$, $\sigma(a) = a^p$. For $Q = F_2/H$ set $C = \langle Q' \rangle_p$ and

$$\begin{aligned} B &= H \cap Z, & N &= (B + Z^{[p]})/Z^{[p]}, & h &= \dim R/N, \\ g &= \dim N/\mathfrak{m}N, & a &= \dim Q/C, & b &= n - a, \\ m &= b - h, & S &= h + g. \end{aligned}$$

In $m = b - h$, the roman m is an integer, while $\mathfrak{m}$ is an ideal. Write C_n for all codimension-n ideals, B_n for those with p-nilpotent C, D_n for those with additionally ordinary-nilpotent Q, and P_n for those with p-nilpotent whole quotient. As in the Section 9, let $\pi_n = D_n/B_n$.

Theorem 10.1 (Rank-two locality costs no linear exponent)**.** *For every fixed finite field there is a constant K_q such that, for all sufficiently large n,*

$$q^{-K_q\sqrt{n}} \le \pi_n \le 1.$$

Consequently

$$\boxed{\frac{P_n}{C_n} = q^{-n/2+O_q(\sqrt{n})}, \qquad \lim_{n\to\infty} -\frac{1}{n}\log_q \frac{P_n}{C_n} = \frac{1}{2}.}$$

These are uniform-ideal statements for the two-generator free source. They neither prove $\pi_n \to 1$ nor transfer this probability rate to uniform unmarked types. No assertion for F_d, $d \geq 3$, is made.

10.2 Near-extremal planar relation ideals have sublinear entropy

For an ideal $I \subset R$ of colength h, put $\mu_0(I) = \dim I/\mathfrak{m}I$. It is the generator number at the origin, even if I has other support. Let $p(h)$ be the partition number.

Proposition 10.2 (A uniform refined-flag bound)**.** *For $g \geq 1$, let $e = h - \binom{g}{2} \geq 0$. Then*

$$\#\{I : \dim R/I = h,\ \mu_0(I) \geq g\} \leq p(h)\, 2^{(1+\sqrt{1+8h})/2}\, q^{2e}.$$

In particular if $h = n/2 + O(\sqrt{n})$ and $g = \sqrt{n} + O(1)$, the number of such relation ideals is $q^{O_q(\sqrt{n})}$.

Proof. Every such I admits an ideal J with $\mathfrak{m}I \subset J \subset I$ and $\dim I/J = g$: choose a codimension-g subspace of $I/\mathfrak{m}I$. Thus it suffices to count these flags. Their complex moduli scheme is the refined Hilbert scheme $H^{[h,h+g]}$. The classical Nakajima–Yoshioka theorem, stated as Theorem 8 in Hsiao–Szenes [9], says this scheme is smooth of dimension $2h - g(g-1) = 2e$ over $\mathbb{C}$. We do not infer finite-field point counts just from this dimension.

First, its tangent dimension at a monomial flag is independent of the field. A tangent vector is a pair

$$\phi_I \in \mathrm{Hom}_R(I, R/I), \qquad \phi_J \in \mathrm{Hom}_R(J, R/J)$$

with $\phi_J(j) \bmod I = \phi_I(j)$ for $j \in J$, and $\phi_J(Xf) = X\phi_I(f)$, $\phi_J(Yf) = Y\phi_I(f)$ in R/J for $f \in I$. These formulas are well defined because $\mathfrak{m}I \subset J$. Use monomial minimal generators and standard-monomial coordinates. Taylor syzygies, the inclusion equations, and the last two equations all equate a single coefficient to another coefficient, or to zero. Multiplication by a monomial is injective on surviving standard terms. Choose one dividing minimal generator whenever evaluating a map on a multiple; Taylor equations ensure independence of that choice. The resulting equation graph has the same rank in every characteristic. Thus the tangent dimension is $2e$ over any field, by the complex smoothness theorem at the same monomial flag.

Second, stratify the flags by their two monomial initial ideals, using one degree-compatible order. The reduced monic Gröbner coefficients for both ideals give a connected positively graded coefficient algebra: choose a single positive weight vector realizing all the finitely many leading-versus-tail comparisons. Buchberger, inclusion and $\mathfrak{m}I \subset J$ equations are homogeneous for these weights. Its tangent space at the monomial flag injects into the $2e$-dimensional flag tangent space. As in Sections 2–5, homogeneous lifts of a cotangent basis generate a connected positively graded algebra, by induction on weight. Hence each stratum has at most q^{2e} rational points. No component-degree or unexplained point-count factor is used.

There are $p(h)$ monomial ideals of colength h. A monomial flag above one is obtained by omitting g of its minimal generator heads, and retaining all their X, Y multiples; there are at most 2^v choices when it has v generators. The staircase bound gives $v \leq (1 + \sqrt{1+8h})/2$. This proves the displayed bound. Finally $\log p(h) = O(\sqrt{h})$ and $e = O(\sqrt{n})$ in the stated range. □

10.3 The effective outer penalty

The compatibility formula of Theorem 8.1 uses a feasible tuple $\alpha \in (R/N)^2$ and a subspace $D \subset N/\mathfrak{m}N$ of rank $r \le 4$, generated by the four centrality errors. Fix any such tuple. Set $J = \mathfrak{m}N + D$, using inverse images in N, so

$$\dim R/J = h + g - r, \qquad \mathfrak{m}N \subset J \subset N.$$

Lemma 10.3 (A local flag in a nilpotent restricted quotient)**.** *Let I be $\mathfrak{m}$-primary of generator number $g \ge 2$, and $\mathfrak{m}I \subset J \subset I$ with $\dim J/\mathfrak{m}I = r$. Suppose there is a two-generated p-nilpotent restricted quotient whose derived algebra is the strongly abelian cyclic module R/J, and whose abelian outer quotient has dimension a_0. Then*

$$a_0 + r \ge T_g := 2\lceil \log_p(g+1) \rceil.$$

Proof. Work in the completed local ring $\mathbb{F}_q[[X, Y]]$. The Hilbert–Burch matrix for I has $g-1$ columns and all its entries in $\mathfrak{m}$. Its maximal minors generate I, so $I \subset \mathfrak{m}^{g-1}$ and $\mathfrak{m}I \subset \mathfrak{m}^g$.

Here is an explicit two-chain reduction that also handles Frobenius. Order the source images so that $\bar{x}$ has maximum nilpotent height α. The quotient of the outer module by $\Omega\bar{x}$ is cyclic, generated by $\bar{y}$, of length $\beta \le \alpha$, and $a_0 = \alpha + \beta$. Write $t^\beta \bar{y} = f(t)\bar{x}$, with $\deg f < \alpha$. Since $t^\alpha \bar{y} = 0$, the t-valuation of f is at least β. As σ is invertible, write $f = t^\beta g(t)$. Replacing y by $y - g(t)x$, with $g(t)x$ lifted as a finite sum of restricted powers of x, gives a second chain of length β disjoint from the first. This is an elementary source substitution with an explicit inverse. It fixes $c = [x, y]$, changes Y by an additive p-polynomial in X, and induces a polynomial coordinate automorphism fixing (X, Y). Thus it preserves g, r, colength and $\mathfrak{m}$-adic order. In particular no skew Smith form, infinite generator series, or characteristic-zero linearization is being used.

Write $P = p^\alpha$, $Q = p^\beta$. The two source powers are elements u, v of the derived module. Since the latter has zero restricted power, centrality of a power with its own source and the restricted adjoint identity give, up to harmless signs,

$$Xu \in J, \quad Yu - X^{P-1} \in J, \quad Xv - Y^{Q-1} \in J, \quad Yv \in J.$$

Setting $Y = 0$ in the second relation and $X = 0$ in the third, and using $J \subset \mathfrak{m}^{g-1}$, shows $P, Q \ge g$. If g is not a p-power this already proves the claim.

Suppose $g = p^k$. If $P = g$, the degree-$(g-1)$ images of J include Xa and $Ya - X^{g-1}$, where a is the degree-$(g-2)$ term of u. There are no lower terms, by $Xu \in J \subset \mathfrak{m}^{g-1}$. These two polynomials cannot both be zero, so $r \ge 1$. The analogous conclusion holds if $Q = g$. If both equal g, let b be the analogous term of v. The four polynomials

$$Xa, \quad Ya - X^{g-1}, \quad Xb - Y^{g-1}, \quad Yb$$

span at least two dimensions. Indeed, if $a = 0$, a one-dimensional span would be generated by X^{g-1}; the last polynomial forces $b = 0$, contradicting the third. If both a, b are nonzero and the span is one-dimensional, its generator is divisible by both X and Y (use Xa and Yb); hence a is divisible by Y, which contradicts $Ya - X^{g-1}$ being in that span. The case $b = 0$ is symmetric. Since the degree-$(g-1)$ image of J has dimension at most r, we obtain $r \ge 2$. Combining the cases gives $\alpha + \beta + r \ge 2k + 2 = T_g$. □

Corollary 10.4 (Non-origin support pays at least one outer unit)**.** *For every feasible compatibility tuple with N not $\mathfrak{m}$-primary, and $g \ge 2$,*

$$\boxed{a + r \ge T_g + 1.}$$

Proof. Make a finite auxiliary quotient by killing J and all positive restricted powers of M, then killing the selected central lifts of the outer basis. The exact central-lift construction in Section 8 shows its dimension is $a+h+g-r$, its outer module is unchanged, and its strongly abelian derived module is R/J.

Let I be the origin-primary component of N, of colength k. The primary decomposition of $R/\mathfrak{m}N$ is its origin part $R/\mathfrak{m}I$, plus the non-origin part of R/N. The space D lies entirely at the origin, because $\mathfrak{m}(N/\mathfrak{m}N) = 0$. Quotient the auxiliary algebra by its non-origin derived module. This is a restricted ideal, since the derived power is zero and all source actions preserve the primary decomposition. The resulting ordinary Lie algebra is nilpotent, with derived module R/J_0, where $\mathfrak{m}I \subset J_0 \subset I$ and $\dim J_0/\mathfrak{m}I = r$. Also $\mu_0(I) = g$.

Apply Section 9's canonical splitting to remove the central invertible outer factor, of dimension t. The remaining quotient is p-nilpotent, has the same derived module R/J_0, and outer dimension $a_0 = a - t$. Here $t \geq 1$: if the original outer module were nilpotent, its extension by a strongly abelian derived module would already be p-nilpotent and could not have non-origin support. Lemma 10.3 gives $a - t + r \geq T_g$, proving the result. □

10.4 A primary comparison model with the same h, g

Lemma 10.5 (A slightly stretched staircase)**.** *Suppose $g \to \infty$, $h = \binom{g}{2} + e$ and $0 \leq e \leq Kg$, for a fixed constant K. There is an $\mathfrak{m}$-primary monomial ideal I_* of colength h, with g minimal generators, and a central truncated outer kernel of dimension a_* satisfying*

$$a_* \leq T_g + 1.$$

For every sufficiently large total dimension n with $m_* = n - a_* - h \geq 0$, this model gives at least

$$q^{(h+g+1)m_*}$$

distinct whole-p-nilpotent ideals of F_2.

Proof. Put $s = g - 1$, write $e = st + u$ with $0 \leq u < s$, and use the strictly decreasing row lengths

$$\lambda_i = s - i + 1 + t + \mathbf{1}_{i \leq u}, \qquad 1 \leq i \leq s.$$

Their sum is h, and their staircase ideal has $s + 1 = g$ generators. Its pure-power exponents are $A = \lambda_1 = s + t + \mathbf{1}_{u>0}$, $B = s$. Choose

$$k_x = \lceil \log_p(A+2) \rceil, \quad k_y = \lceil \log_p(B+2) \rceil, \quad a_* = k_x + k_y.$$

Since $t = O_K(1)$, this is at most $2\lceil \log_p(g+1) \rceil + 1$ for sufficiently large g. Moreover $X^{p^{k_x}-1}, Y^{p^{k_y}-1} \in \mathfrak{m}I_*$. Thus the discarded pure outer towers are central modulo $\langle \mathfrak{m}I_* \rangle_p$, and have a nilpotent outer quotient.

For every prescribed-head nilpotent inner kernel of dimension $b_* = n - a_*$, take the zero head corrections of these two outer generators and arbitrary tails in $C^{[p]}$. There are q^{2m_*} such lifts. The nilpotent head count of rank $h + g$ is at least $q^{(h+g-1)m_*}$. The exact compatibility theorem ensures these are distinct ideals with whole-p-nilpotent quotient. Multiplying proves the lower bound. □

10.5 Proof of the locality cost and the full rate

Proof of Theorem 10.1. The rank-two concentration argument of Theorem 7.2 applies to the inner-nilpotent ensemble B_n: its denominator is at least P_n, and its numerator is bounded by

all ideals. By increasing the bulk constant, for any fixed $L > 0$ all but a fraction q^{-Ln} of B_n satisfy

$$h = n/2 + O_q(\sqrt{n}), \quad g = \sqrt{n} + O_q(1), \quad a = \log_p n + O_q(1).$$

Fix constants once so that the exceptional fraction tends to zero. In this bulk $e = h - \binom{g}{2} = O_q(\sqrt{n})$, and $S + 1 - m = 2h + g + a + 1 - n = O_q(\sqrt{n})$.

For a fixed N, O, feasibility of a head correction is an affine linear system. Its homogeneous solution space is $(W^{\mathfrak{m}})^2$. Since $W = R/N$ is cyclic and $\mu_0(N) = g \geq 2$, the planar Koszul Euler characteristic gives $\dim W^{\mathfrak{m}} = g - 1$. Therefore there are at most $q^{2(g-1)} = q^{O_q(\sqrt{n})}$ feasible tuples. For a feasible tuple of rank $r \leq 4$, its nilpotent contribution is at most

$$\Theta_q q^F, \qquad F = (S - r + 1)(n - a - h).$$

If N is nonlocal, Corollary 10.4 and Lemma 10.5 give a primary model with the same h, g and $a_* \leq T_g + 1 \leq a + r$. Its exponent $F_* = (S + 1)(n - a_* - h)$ obeys

$$F - F_* = (S + 1)(a_* - a) - rm \leq r(S + 1 - m) = O_q(\sqrt{n}).$$

Consequently each such tuple contributes at most $q^{O_q(\sqrt{n})} P_n$.

There are only $q^{O_q(\sqrt{n})}$ relation ideals in the bulk, by Proposition 10.2; outer choices cost $q^{2a+O_q(1)} = q^{O_q(\log n)}$, and parameter summation is polynomial. The bound for feasible tuples was already recorded. Thus the whole nonlocal bulk contributes at most $q^{O_q(\sqrt{n})} P_n$. All local ideals, including those outside the bulk, are counted by $D_n \leq 2P_n$ for large n, by the exact convolution and smoothness theorem in Section 9. Absorbing the exceptional fraction of B_n gives

$$B_n \leq q^{O_q(\sqrt{n})} P_n.$$

Since $P_n \leq D_n \leq B_n$, this proves the claimed bound on π_n.

Finally the conditional inner-nilpotence ratio of Section 8 is uniformly $q^{-m+O_q(1)}$. In a rank-two all-ideal bulk, $m = n/2 + O_q(\sqrt{n})$; choose its exceptional fraction to be q^{-Ln} with $L > 1/2$. Averaging gives $B_n/C_n = q^{-n/2+O_q(\sqrt{n})}$. Combining with $q^{-O_q(\sqrt{n})} \leq P_n/B_n \leq 1$ proves the full rate and completes the argument. □

11 Further questions and reproducibility

11.1 Beyond rank two

The rank-two theorem determines a linear exponential rate. It allows π_n to decrease on a sublinear exponential scale. Its limit and its $\sqrt{n}$ correction remain to be determined. For three or more source generators, the planar refined-flag bound, the cyclic socle identity, and the two-chain outer comparison each require a replacement.

The fine probability statements use uniform ideals. The exact type convolution in Corollary 9.4 remains valid, but detailed laws under uniform unmarked isomorphism types require control of marking multiplicities at a finer scale than the factor q^{dn} used for coarse growth. This is a separate counting problem.

11.2 Exact controls

The source archive includes the original head and semilinear-block checks and five self-contained programs in `verification/typical_structure/`. They reconstruct finite kernels, centrality equations, partition counts, and the refined-flag tangent systems using exact arithmetic. The coverage includes genuine Frobenius over $\mathbb{F}_4$: 24 prescribed-head cases; 24 independent outer-compatibility models; exact partition and condensation identities; 84

marked/type toral convolutions; and the rank-two locality controls (2710 monomial flags, 7596 primary comparison models, 20645 boundary-rank cases, 1620 exponent comparisons, and 6992 semilinear two-chain reductions). The general proofs specify the arbitrary-size statements; the finite controls check their algebraic interfaces. The accompanying README gives one command per program and the corresponding report files.

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